Roku Game Engine — User Manual

Production Alpha · v0.1.0 · Desktop authoring and native Roku runtime

Create scenes, artwork, menus, animation, audio and BrightScript behaviors; then build, test and deploy a native Roku game.

This manual describes the active application in this repository. Reference tables are generated from its current schema and visible controls. Screenshots show the included Collect the Stars project in the earlier Dark theme. Mid-tone is now the default for new preferences. Docks can move, so your arrangement may differ. Explanatory diagrams are labeled illustrations, not screenshots of additional features.

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Reference coverage: 35 creatable nodes · 12 components · 505 node property entries · 230 static controls · 202 bound engine signatures · 15 figures.

1. What the engine is

Roku Game Engine is a desktop 2D editor built around a scene tree. Arrange visual and logical nodes, change their properties, attach reusable components, and write BrightScript when built-in behaviors are not enough. The result is a native Roku package containing BrightScript, assets and a manifest. The authoring canvas uses Konva; exported games use the canonical runtime under source/engine.

It supports remote-controlled arcade games, simple scrolling games, puzzles, menus, HUDs and prototypes. Its active tools include camera helpers, tile painting, image slicing, animation, AABB physics, audio authoring, persistent options/high scores, simulator testing and developer-device deployment.

The scene canvas and camera preview are authoring views. Build & Test executes the exported native package in the simulator. This is an alpha: verify final input, audio, memory and performance on the Roku models you intend to support.

Term	Meaning
Project	A folder containing project.settings.json, scenes, artwork, audio, fonts and scripts.
Scene	A saved .rkscene document containing a root and a hierarchy of nodes.
Node	An object in the hierarchy: sprite, camera, menu control, timer, path or other helper.
Component	A behavior attached to a node: movement, collision, audio, animation or a script.
Resource	A reusable image, sprite sheet, frame library, tileset, style, font or audio clip.
Inspector	The property editor for the selected item.
Runtime	The BrightScript engine executing the exported game.
res://	Project-relative path, such as res://images/player.png.
pkg:/	The resource prefix inside the native package.

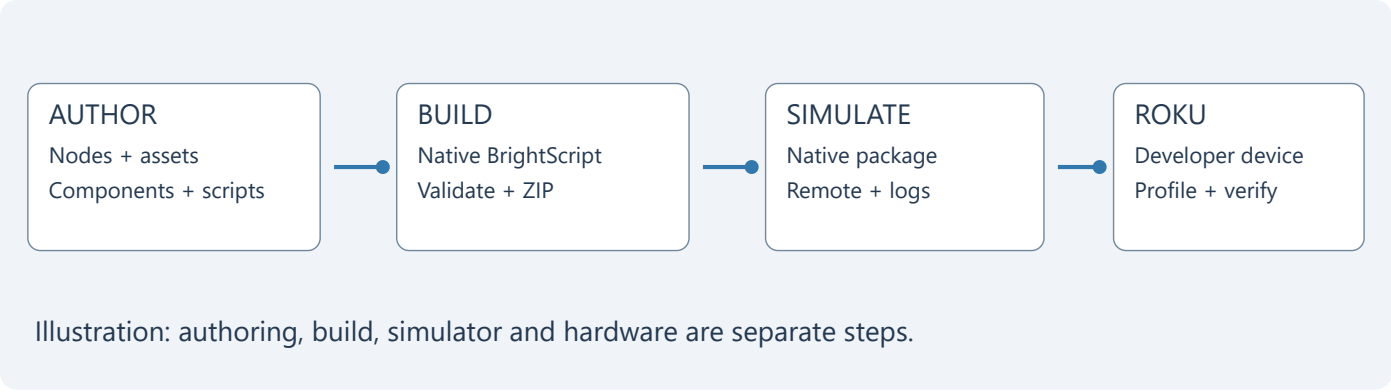


Figure 1. Build pipeline. A visible authoring preview is not a native execution test.

2. Install, launch and first build

Install Node.js 22.12 or newer and Git. Open PowerShell in the repository folder. Initial dependency and simulator setup require downloads.

```
npm.cmd ci
npm.cmd run simulator:setup
npm.cmd start
```

Use npm.cmd start on subsequent launches. In other shells, npm can replace npm.cmd. Do not double-click index.html: filesystem, native builds and device operations depend on Electron.

In the project manager, select a project to inspect its details. Open its folder rather than an individual asset. Search/sort controls organize the project list. Create a project for a new game; open samples/collect-the-stars to follow this guide.

Create makes a project folder; Import registers an existing one; Scan Folder Projects searches a local folder, not the Roku network. The selected project's Edit opens it; Export builds it; Rename changes its folder; Duplicate copies it; Manage Tags edits comma-separated tags; Remove removes the manager entry without deleting the folder. Project-manager Settings controls default list sorting and recent-project limit, not game input or the editor theme.

1. Open samples/collect-the-stars, then CollectStars.rkscene.
2. Select player in the Scene tree and inspect its transform/components.
3. Choose the startup scene in the top toolbar.
4. Click Build & Test or press F5. Wait for export, validation and simulator startup.
5. Use the simulator remote's Select/OK action to start; move Right to collect stars; use Select to restart after winning.
6. Change a label or position, build again, and save your authored scene separately.

Build & Test includes current in-memory scene edits. It is not a substitute for saving the scene/project. Scripts have a separate save state; builds save edited scripts through the script editor before compiling.

3. Workspace and docking

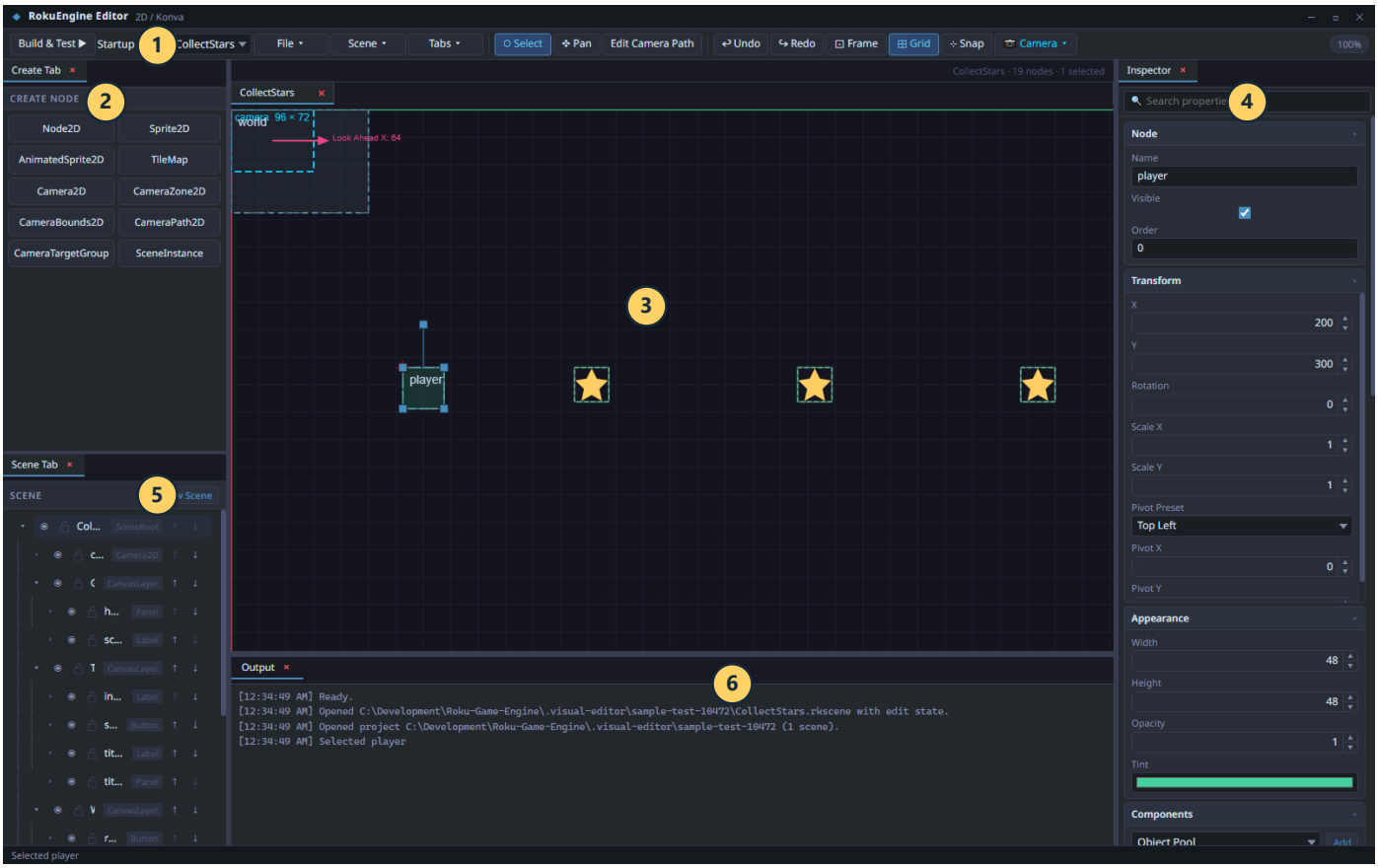


Figure 2. Workspace screenshot (Dark theme): 1 toolbar; 2 Create/tools; 3 world canvas; 4 Inspector; 5 Scene tree; 6 Output.

The numbered markers locate the toolbar, left tools, world canvas, Inspector, Scene tree and Output. The center has open-scene tabs; the right Inspector follows the active selection. Specialist tools appear in docks. Use Tabs to reveal a hidden tool, drag dividers to resize areas, and Tabs → Default to restore the default arrangement. Closing a dock tab hides the tool; it does not delete its data. Return to dock exits the UI designer's expanded arrangement.

Select nodes through the tree or canvas. Selection handles describe the editing rectangle. Camera frames, path lines, trigger outlines and marker crosses are guides, not artwork exported into the game. Eye/visibility controls on actual nodes can affect the game; guide toggles in the Camera menu only change authoring overlays.

World nodes live in camera-controlled space. UI nodes normally belong beneath CanvasLayer and use screen-space layout. Place the HUD under CanvasLayer rather than under the player sprite so world movement does not become menu placement.

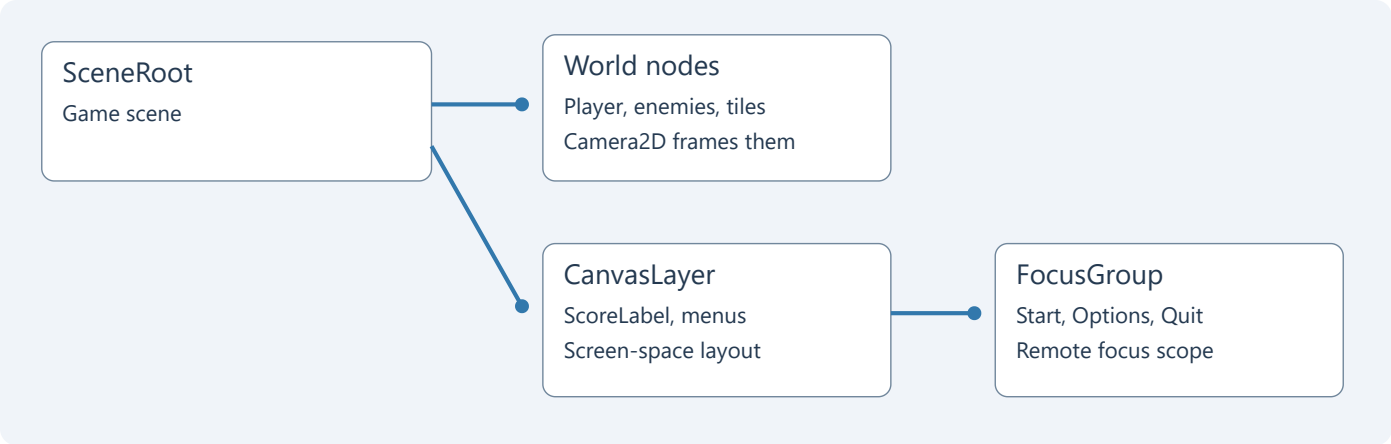


Figure 3. Illustration: separate world content from screen-space HUD/menu hierarchy.

4. Icons, buttons and toolbar

Tooltips are the quickest reminder of a symbol's meaning. A plus usually adds an item in the current list. A tab cross closes/hides the tab; a delete/trash control removes selected data. A triangle usually plays/previews. A circular arrow refreshes that tool; it does not necessarily restart the game.

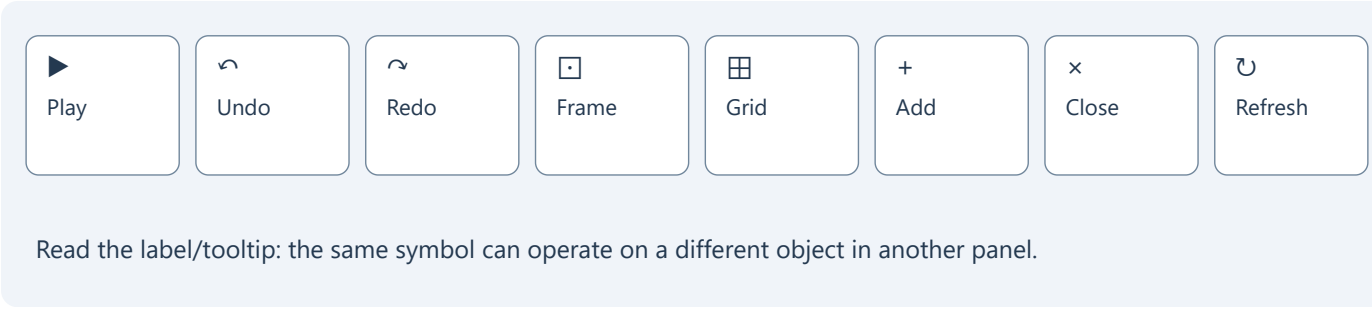


Figure 4. Symbol guide. Text labels are the primary identifiers when fonts render an icon differently.

Tree and FileSystem glyphs

Glyph / kind	Meaning
⬡ · CanvasLayer	UI element identifier. Select its row to edit the element; see the node catalog.
⊞ · Control	UI element identifier. Select its row to edit the element; see the node catalog.
▭ · Panel	UI element identifier. Select its row to edit the element; see the node catalog.
T · Label	UI element identifier. Select its row to edit the element; see the node catalog.
▢ · TextureRect	UI element identifier. Select its row to edit the element; see the node catalog.
⊙ · Button	UI element identifier. Select its row to edit the element; see the node catalog.
⊟ · NineSlicePanel	UI element identifier. Select its row to edit the element; see the node catalog.
≡ · MenuList	UI element identifier. Select its row to edit the element; see the node catalog.

Glyph / kind	Meaning
▬ · ProgressBar	UI element identifier. Select its row to edit the element; see the node catalog.
📂 · directory	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.
🎬 · scene	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.
🖼 · image	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.
🎵 · audio	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.
📦 · component	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.
📜 · brightscript	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.
📜 · script	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.
🧩 · json	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.
📄 · file	FileSystem identifier. The icon identifies the resource kind; its filename/path is the reliable reference.

Icon/label	Action and consequence
Build & Test / play triangle	Export, validate, package and run the native game. Shortcut: F5. Read compiler diagnostics if it fails.
Startup scene	Choose the entry scene for the next build; distinct from the active editing tab.
Select / selection symbol	Select and manipulate scene objects.
Pan / four-way symbol	Move the editing view without moving game nodes or the runtime camera.
Edit Path	Edit points on selected Path2D/CameraPath2D rather than moving the whole node.

Icon/label	Action and consequence
Undo / curved left arrow	Undo a supported edit in the current editor. Text has its own history.
Redo / curved right arrow	Reapply an undone edit. New edits can replace the redo branch.
Frame / framing box	Bring selection into view; node coordinates are unchanged.
Grid / square grid	Toggle the editing grid, which is not exported artwork.
Snap	Toggle snapping for supported edits. Check snap settings when motion seems constrained.
Camera dropdown	Show/hide guides and open camera navigation/preview tools.
Eye in a tree	Toggle the node's authored visibility.
Lock in a tree	Protect/unprotect an item from accidental editing interaction.
Disclosure triangle	Expand/collapse a tree branch or property section without disabling it.
Plus	Add an item appropriate to the current resource list or node context.
Tab cross	Hide/close a tab; it is not node deletion.
Refresh / circular arrow	Refresh that panel's preview or data.
Fit / 100% / percentage	Change authoring-preview magnification, not Camera2D runtime zoom.
Gear / Options	Open configuration for its current context. Project Settings is File → Settings.

File menu

Create Project starts a new project; Open Project selects a folder; Back to Manager returns to project selection. Save Project saves project work; Save As saves through the project-copy workflow. Export Game produces a Roku package. Build & Test in Roku Simulator runs the native build path. Sideload to Roku opens the developer-device dialog. Settings opens project/input/screen configuration and editor appearance. Quit exits.

The Scene panel search finds editable nodes by name, type or ID, ignoring case. Combine words to narrow results. Matching nodes appear with their ancestors even when those branches were collapsed. Enter selects the first match; Escape or the clear button restores the full tree. Select a node to drag it from inside its selection rectangle, including when guide overlays cover its artwork. Locked nodes remain fixed.

Scene menu

New Scene creates a scene; Open Scene opens one; Rename Scene changes its name. Save Scene and Save Scene As persist it. Reload Saved Scene replaces current memory state with the saved version: preserve edits first. New Inherited Scene derives from a base; Rebase Inherited Scene updates from its base; Revert Inherited To Base removes its overrides through that workflow. Back up before intentionally discarding overrides. Close Scene/Close All Scenes close editing tabs, not runtime scenes.

5. Nodes, transforms and Inspector

Create adds a node from the available types. Select the intended parent first and inspect the resulting hierarchy. The tree/context menu exposes available hierarchy, duplication and deletion operations. Names help humans; IDs are stable references. Avoid changing IDs used in scripts and avoid ambiguous duplicate names when target settings resolve by name.

X/Y are parent-local coordinates, positive right/down. Rotation is degrees. Scale multiplies the node and descendants; negative scale flips supported artwork. Pivot X/Y are local pixel units describing the rotation/scale origin. Pivot Preset is a convenient origin choice. Width/height size the rectangle; scale changes its transformation. Z/order determines relevant sibling drawing order. Opacity is normally 0–1 and inherited. Node tint/colors belong to the game, independently of editor appearance.

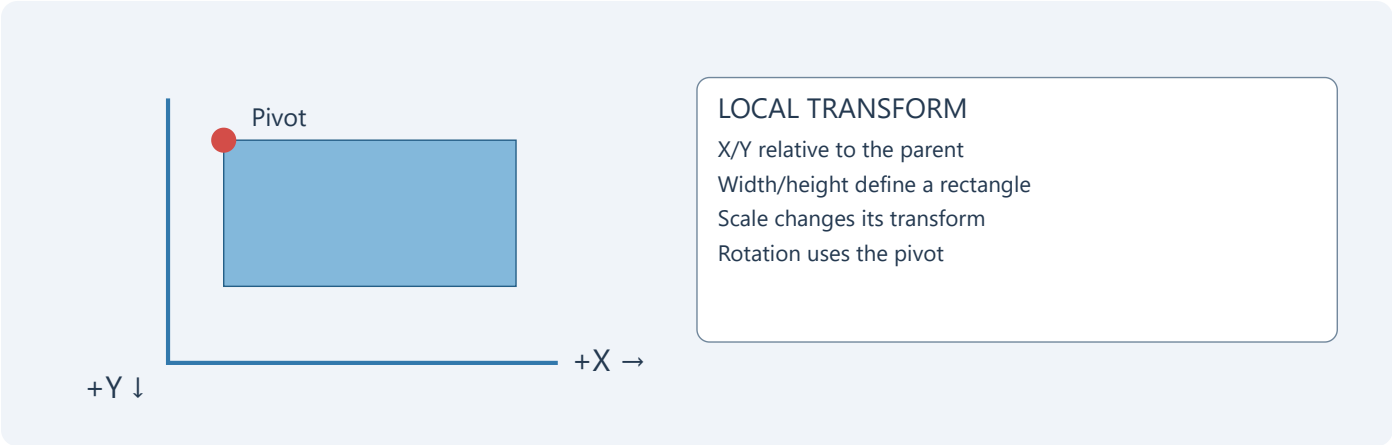


Figure 5. Illustration: parent-local coordinates, local rectangle and pivot. Positive Y points down.

Inspector search finds properties. Disclosure arrows open sections. Numeric steppers adjust values; typing sets exact values. Checkboxes are boolean toggles; dropdowns choose modes; resource pickers assign project assets. An empty target/resource normally means unassigned, not an automatic search for artwork.

SceneInstance references a reusable scene: use it for enemies, menu sections and prefabs. The source is the template, while per-instance overrides customize it. Editable children and selected override determine what you are changing. Fix missing source references rather than treating a placeholder as the source. Runtime scene loads use fresh authored copies; previous playthrough mutations do not replace the saved template.

Key/gesture	Context and result
F5	Build & Test.
Arrow keys / Shift + Arrow	Nudge world selection by one unit / grid step when not editing text.
Ctrl/Cmd + D	Duplicate selected world nodes outside text editing.
Delete / Backspace	Delete world selection; in path mode, delete selected points.

Key/gesture	Context and result
Escape	Close applicable menus or leave/cancel applicable point editing.
O / E / P on world viewport	Tile picker / erase / paint operations; context-sensitive.
Ctrl/Cmd + S in Scripts	Save the active script.
Ctrl/Cmd + F in Scripts	Find text; Find next advances the search.
Tab in Scripts	Insert two-space indentation.
Wheel in UI preview	Pan; Ctrl + wheel zooms; Alt + drag pans.

6. Images, sprites and slicing

Sprite2D displays an image. Assign a project-relative path. Contain shows the whole image within the rectangle and may leave space; Cover fills the rectangle while cropping overflow; Stretch fills it by changing aspect ratio. PNG is predictable raster artwork. The native build can rasterize referenced SVG to PNG; external SVG resources are rejected.

Sprites manages sprite sheets. Upload an image, set Tile W/H, Padding and Offset X/Y, then select cells. Grid toggles the overlay. Search/sort/category filters organize sheets. A SpriteFrames library groups frame indices into named animation clips. Changing sheet geometry can change which pixels an existing frame index selects.

Sprite Slicer procedure

1. Open Tabs → Sprite Slicer. Import image or Use selected element image.
2. Enter cell dimensions, spacing and offsets; choose Slice grid. Complete cells are generated in row order.
3. Select cells; Ctrl/Shift-click toggles multi-selection. For irregular atlases choose Draw rectangle and drag a region. Zoom/scroll as needed.
4. Name the slices and Save selected regions. Batch names are numbered; the source image is preserved.
5. Select a saved region and scene element, then Apply region to selected element. Existing size, position, pivot and script connections remain.
6. Create from region instead makes Sprite2D, TextureRect, Button or Panel at natural size with a top-left pivot.
7. Save the scene and resource state.

UI borders preserve edge pixels as a button/panel resizes. Left/top/right/bottom borders must leave a center. Zero borders give a simple crop. Edit UI borders / states opens detailed atlas styling. Slicing creates reusable regions, not animation clips; author those in Sprites/Sprite Frames.

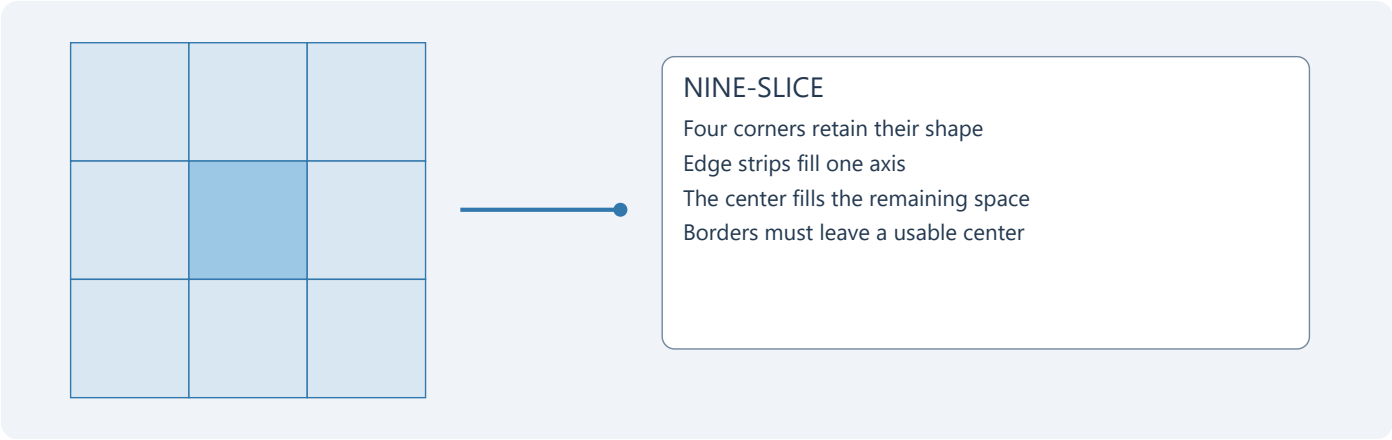


Figure 6. Illustration: nine-slice borders resize surfaces without stretching their corner details.

7. Animation and graphs

AnimatedSprite2D needs a SpriteFrames library and valid animation name. Frame supports manual/non-playing selection. Playing advances frames; Speed Scale changes playback rate. AnimationPlayerComponent provides component playback and graph integration.

Set graphPath on that component to a project JSON graph. The sprite's spriteFramesId points at an imported library; graph clips can override it. A graph clip name selects library metadata, while its frames are sheet indices. Parameters drive transitions. Duration fades the previous state image into the new state; this is an image crossfade, not skeletal blending.

```
{
  "id":"hero", "defaultState":"idle",
  "parameters":{"Speed":{"type":"float","defaultValue":0}},
  "states":[{"id":"idle","clipId":"idleClip"},{"id":"run","clipId":"runClip"}],
  "clips":[
    {"id":"idleClip","name":"Idle","frames":[0,1],"fps":4,"loop":true},
    {"id":"runClip","name":"Run","frames":[2,3,4,5],"fps":10,"loop":true,
     "events":[{"frame":2,"name":"footstep"}]}
  ],
  "transitions":[
    {"from":"idle","to":"run","duration":0.15,
     "conditions":[{"param":"Speed","op":">","value":0}]},
    {"from":"run","to":"idle","duration":0.15,
     "conditions":[{"param":"Speed","op":"==","value":0}]}
  ]
}
```

Call context.engine.SetAnimationParameter(context.node.id, "Speed", 2.0) from an attached script. Initial graphParameters accepts JSON such as {"Speed":0}. Set trigger parameters true; evaluation clears them. AnimationEvent supplies authored frame events; AnimationStateChanged identifies transitions. Clearing graphPath restores ordinary animation. Catch-up events are bounded to eight cycles and 64 events per update.

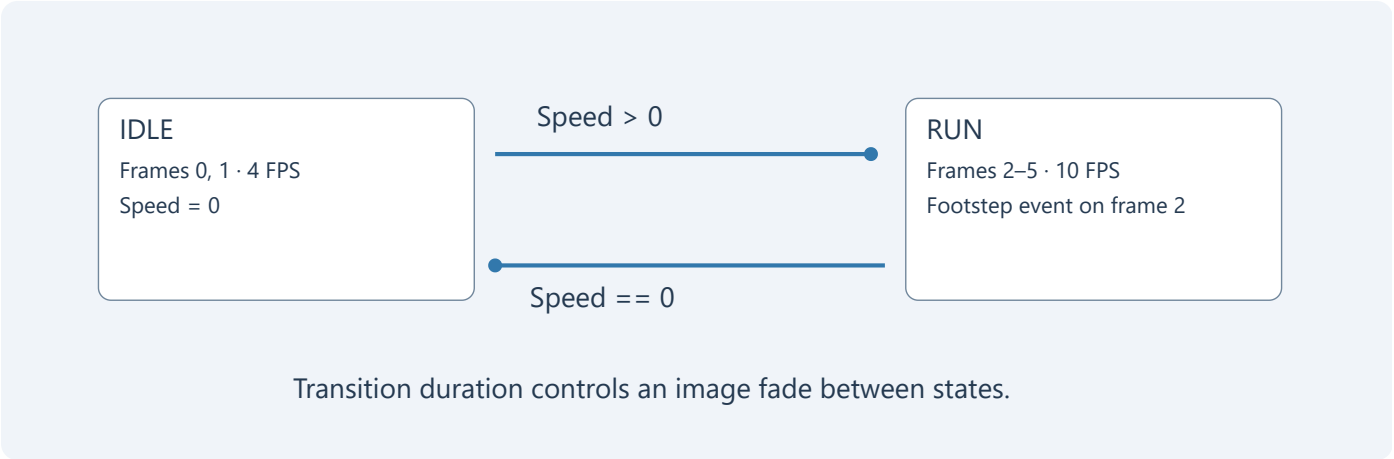


Figure 7. Illustration: parameter-driven animation state changes; clips still need valid frame-library metadata.

8. Tilemaps, layers and terrain

Import the sheet, create a tileset, then assign it to TileMap. Map Width/Height are cell counts; Tile Width/Height are cell dimensions. The node transform positions the map. Choose the active layer before painting. Layers separate background/decorations/gameplay content.

Tool/mode	Purpose
Paint / Rect / Line	Place the selected tile or brush cell-by-cell, as a rectangle or along a line.
Erase	Remove cells from the active layer.
Fill	Fill a connected area under the current mode/brush rules.
Pick	Sample a placed tile into the current brush.
Tiles / Terrain / Collision	Edit explicit tile identities / terrain rules / collision-related data.
+ Layer	Add a layer to the map.
Sync Screen	Synchronize map dimensions using the current screen setup.
Clear	Clear the relevant map/layer data; verify the active context first.
Rules / Collision overlays	Reveal terrain-rule or collision metadata on the tileset.
Select w/ Rules	Select tiles that have terrain rules.
Clear Selection	Clear editing selection without deleting the tiles.

Terrain sets organize terrain types and neighbor/socket rules. Define terrain types, assign appropriate rule data and variants to tiles, then paint terrain. Probability/randomized interiors choose among eligible alternatives; they do not create missing art. Save the scene/resource data after tileset changes.

Tile Opts includes collision/physics shape tools: Rect draws a rectangle; Poly clicks points and double-click closes; Select selects/moves shapes; Full Tile inserts a tile-sized box; Clear All removes those shapes. Native physics uses enclosing AABBs for polygon/compound data, not polygon rigid-body dynamics. If painting fails, check map, tileset, active layer, brush, visibility and mode before repainting a level.

9. Cameras and overlays

Set Current and Enabled on the Camera2D that should drive the game. Follow Target identifies the tracked node. Without a follow target, use a static position. Follow Lerp/Smooth Follow control response; Offset changes framing. A deadzone allows movement before following. Look Ahead anticipates target direction. Runtime camera Zoom differs from editor viewport zoom.

CameraBounds2D/camera bounds restrict travel. CameraZone2D applies regional behavior: the highest-priority enabled containing zone wins, Blend Time controls transition timing and Room Lock centers a room. Target groups frame referenced nodes with padding and zoom limits. CameraPath2D follows authored points; Auto Play starts automatically. Speed takes precedence over Duration; a speed below 1 uses duration. Camera paths support Linear/Smooth/Cubic interpolation; gameplay Path2D is separate and uses straight segments.

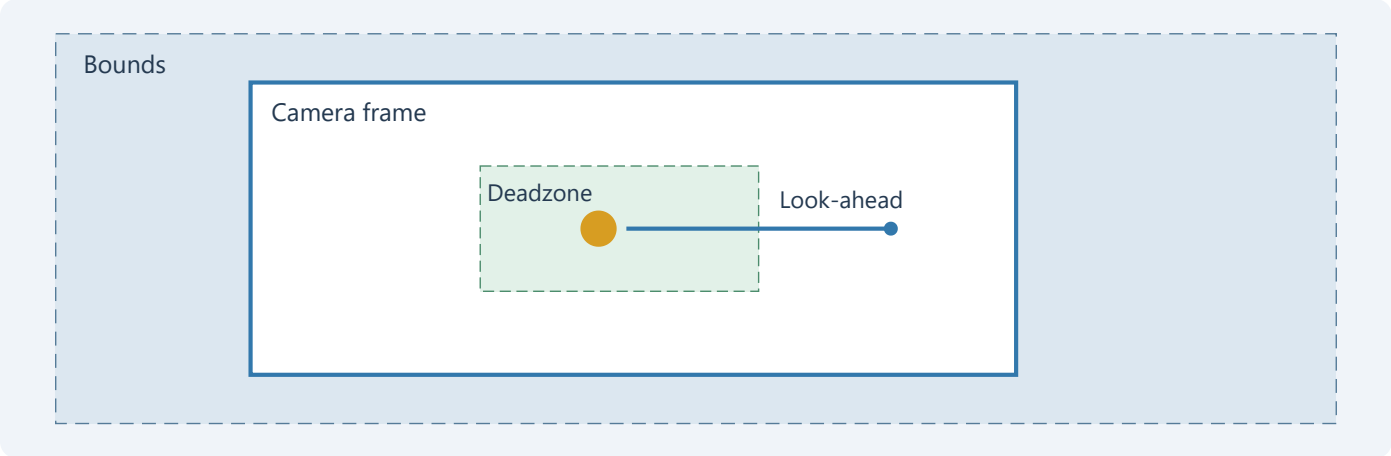


Figure 8. Illustration: bounds limit travel, the frame shows the view, deadzone tolerates target movement, and look-ahead shifts framing.

The Camera menu controls frame, deadzone, look-ahead, bounds, labels, zones, paths and target-group guides. These are authoring overlays. Focus/Frame/Jump To Camera navigate the editing view. Toggle Preview Panel opens the camera preview. Select a camera, Fit or choose magnification, pause/resume and refresh. Play Path/Stop preview the selected camera path; they do not record a movie or replace a native run.

Native camera view rotation is not implemented. A rotated camera outline does not rotate the exported view. Sprite rotation is supported separately. Consult build warnings before depending on unsupported UI/camera transforms.

10. UI Screens and layout

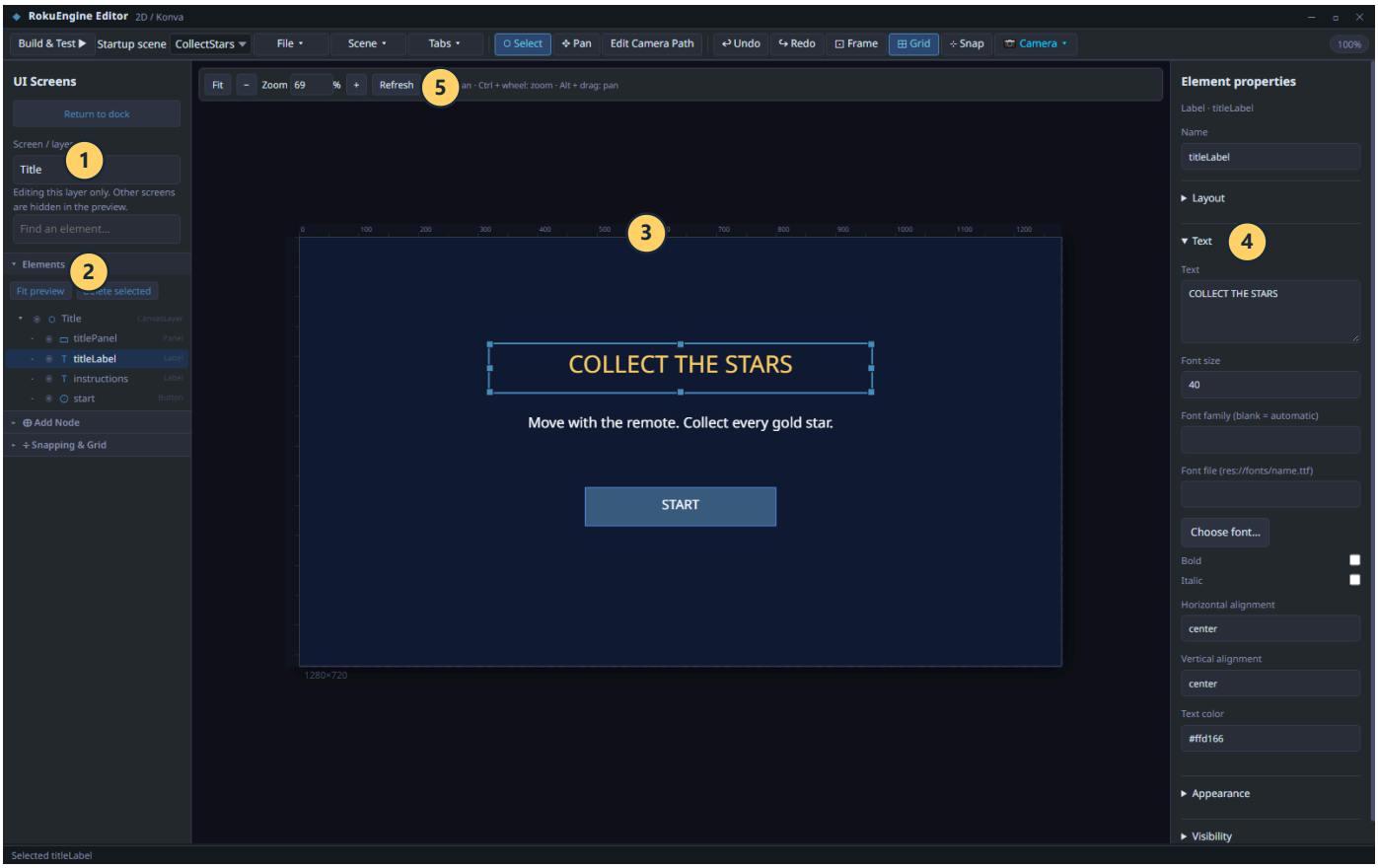


Figure 9. UI Screens screenshot (Dark theme): 1 screen/layer; 2 Elements tree; 3 selected screen canvas; 4 element properties; 5 Fit/zoom/refresh controls.

Open Tabs → UI Screens. Choose a screen/layer to isolate it. Select a control in Elements and edit Element properties. Fit, minus/plus and percentage change magnification; Refresh redraws; Return to dock restores docking. Other screens can be hidden in the preview without being removed from the project.

Begin with CanvasLayer, then add Control, Panel, Label, Button and other controls beneath it. Anchors are fractions of parent dimensions; margins are pixel offsets from those anchors. Right/bottom anchors of 1 follow those edges. Stretch presets choose matching anchor pairs. Alignment tools arrange selections. Container-owned children can have authored positions overridden by layout.

Layout node	Native behavior
Control	General rectangle/group; clipContent constrains descendants.
HBoxContainer	Visible children in a row with separation and begin/center/end alignment.
VBoxContainer	Visible children in a column with separation and alignment.

Layout node	Native behavior
MarginContainer	Children fill its padded inner rectangle.
ScrollContainer	Offset the content and clip descendants to the viewport; change scrollOffsetX/Y by property/script.

Oversized box children shrink proportionally to fit. Check text and buttons at the intended screen resolution. UI supports anchors, inherited scale, clipping, textures and opacity; rotated UI and affine shear are outside the native renderer.

Label stores text, size, color and alignment. UI typography controls expose available font file/family and style options. Choose font selects a project font; resource paths usually look like res://fonts/name.ttf. Build/target support still matters. Large type in a short rectangle can clip.

TextureRect displays fitted artwork; Panel is a colored/styled rectangle. NineSlicePanel preserves textured edges/corners. ProgressBar maps Value between Min/Max with an optional label and style regions: update Value from gameplay rather than resizing its rectangle each frame.

11. Buttons, focus and actions

Button, CircleButton, TriangleButton and OvalButton share state colors, text, focus and action settings. The shape affects drawing; navigation remains remote-driven. Focused Color identifies current focus; Pressed Color indicates activation. Default Focus provides an initial choice where applicable. Disabled controls cannot be activated normally.

Explicit Focus Up/Down/Left/Right links resolve ambiguous spatial navigation. FocusGroup traps related menu controls; Initial Focus starts selection, Wrap moves around edges, Restore Focus remembers a valid prior control. Disabling the group removes descendants from navigation.

MenuList items can be strings or objects with label, action and disabled. Up/down changes selectedIndex and skips disabled entries; the visible row window follows selection. OK uses the item's action or the node action. Set wrap false to stop at the ends. MenuSelectionChanged/MenuItemSelected hooks belong on the MenuList owner.

```
["Continue", {"label":"Start","action":"scene:Game"},
 {"label":"Locked","disabled":true}]
```

Action string	Runtime effect
scene:Game / push:Pause	Change scene / push a scene through scene-stack behavior.
screen:Title / pushscreen:Pause	Change / push a UI screen.
popscreen	Pop the UI screen stack.
transition_screen:Game / transition_scene:Game	Change screen/scene using the default transition.
custom:StartGame	Deliver a custom action/event for scripts.
option_cycle:musicVolume	Cycle a defined option.
option_set:difficulty=hard	Assign a defined option.
options_reset	Restore option defaults.

Names must match existing scenes/screens/options. Connect selected button in Scripts replaces the old action with a generated custom action. Save both script source and scene attachment/action data.

12. UI assets, transitions, options and scores

UI Assets Atlas stores reusable nine-slice styles. Pick Image selects a source. Slice borders protect corners/edges; Center/Edge modes choose how other areas draw. Button styles can have state regions; progress styles can have background/fill/thumb regions. Save Style persists the definition; Delete removes it, so check its users first.

Transitions authors presets: plus creates, type/easing/timing configure, Preview auditions, Duplicate copies and Delete removes. Runtime changes must reference a preset or supported action/API. A preview does not prove device performance.

Options defines persistent settings. Add built-in options inserts the supplied definitions; plus creates custom definitions; delete removes the selected definition. Each key/type/default provides data for actions and UI bindings. High Scores creates named local score tables, not a network leaderboard. Configure limits and bind entry/display UI. Device registry data persists separately from the editor definition; deleting a definition is not an explicit storage clear.

Templates generate editable starter UI structures. Choose one, review startup options and Create. Adapt labels/actions to existing scenes. A template is a starting point, not a complete game controller.

13. Gameplay helpers and pooling

Marker2D is a named position for starts, spawns or checkpoints. Timer emits Timeout in seconds: Autostart starts it, One Shot ends after one firing, Paused stops that timer. Process When Paused permits real-time advancement while the scene is paused.

TriggerZone2D is a trigger without a separate Collider2D. Contact Method selects project default, AABB box or circle; Circle Radius 0 fits half the smaller dimension. Set Layer/Mask and attach collision hooks. Path2D stores local points. PathFollow2D references a path or its Path2D parent; put artwork beneath it. Speed is path-local units/second, Progress sets/seeks distance, Loop repeats, Rotates follows its tangent. Negative speed is supported.

Parallax2D changes descendants' camera-relative motion. Scroll Scale 0 stays fixed onscreen; 1 behaves like world motion. Autoscroll adds units/second. Positive repeat dimensions repeat artwork; zero disables that axis. Repeated artwork does not duplicate gameplay collision.

VisibilityEnabler2D suspends target processing/physics outside the camera plus Margin. It does not rewrite authored visibility/enabled flags. Multiple regions keep the target active if any is onscreen. Spawner2D clones an existing template hierarchy or acquires a pool item. Spawn Points cycle marker references; an empty list uses the spawner origin. Interval is seconds per burst, Spawn Count is requests per burst, Max Active caps live roots.

ObjectPoolComponent preallocates reusable objects. PoolAcquire returns an item or invalid if exhausted; PoolRelease suspends it until reused. Merely hiding a node is not pool release. See samples/gameplay-helpers for the helper types and script hooks. Repeating timers dispatch at most 32 callbacks per update; automatic spawners issue at most one burst of up to 64 objects; parallax drawing caps copies per descendant.

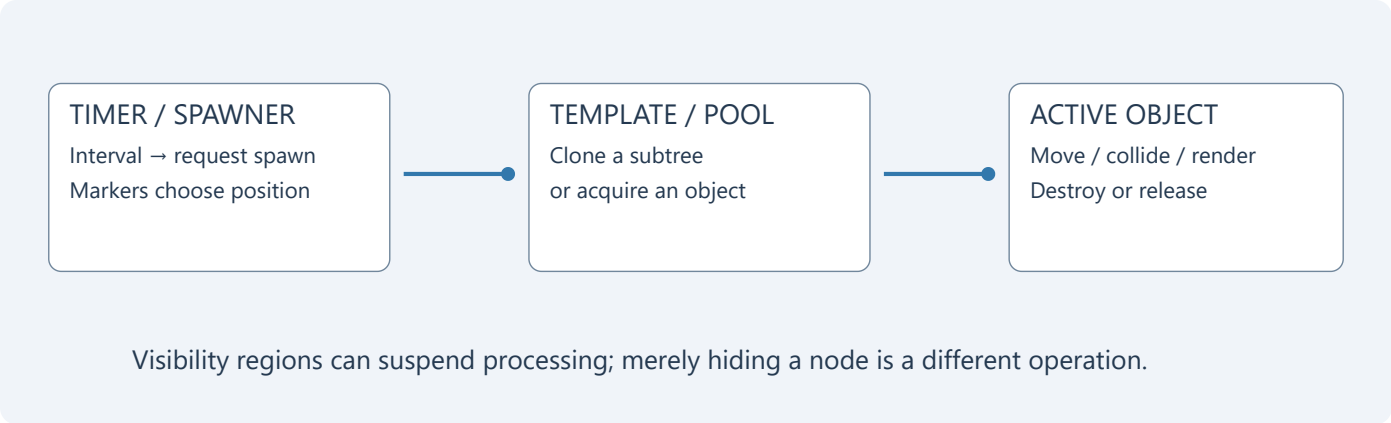


Figure 10. Illustration: bounded spawning and reusable pooled objects.

14. Components, input and physics

Add a component through the node Inspector. Enabled toggles the behavior; Priority orders execution within its phase. Avoid competing movement behaviors unless their combination is intentional. Move adds velocity, Spin rotates, Bob adds a sine offset, input components move from remote actions, Enemy AI supplies selected movement patterns, Object Pool prepares reusable objects, and animation/audio/script components connect those systems.

Collider2DComponent supplies axis-aligned dimensions/offsets, trigger state and collision Layer/Mask. RigidBody2DComponent supplies body type, velocity, acceleration, gravity, friction, bounce and mass. Dynamic bodies advance through physics; choose static/kinematic when appropriate. Lock Rotation prevents SpinComponent rotation through the rigid-body path.

For triggers, Contact Method selects project default, AABB (box bounds) or Circle (distance checks). File → Settings → Physics sets the project default; existing projects default to AABB. Circle Radius 0 uses half the smaller Width/Height; a positive value sets a local radius. The circle center is the rectangle midpoint plus offsets. World radius uses the largest transform stretch, keeping a conservative circle under nonuniform scale. Circle mode ignores custom Physics Shapes. Solid colliders always use AABB physics.

Circle-circle contacts compare squared distances without a per-pair square root; mixed circle-box contacts use the closest box point. Both choices retain spatial buckets, layer masks and Enter/Stay/Exit hooks. Neither guarantees higher frame rate: scene traversal, transforms, scripts and drawing also cost time. Rebuild after changing project settings.

Solid movement sweeps against AABBs and resolves position/velocity; collision hooks report contacts. Trigger checks use fixed-step overlap, so very fast objects can cross thin triggers. Use `engine.QuerySegment(x1, y1, x2, y2, radius, skipId, colliders)` for swept projectile queries against the selected box/circle contacts; it returns ordered hit records with nodeId, fraction, x, y and collider. Radius defaults to 0, skipId to an empty string, and an omitted collider list builds a native snapshot. Polygon/rotated box shapes use enclosing AABBs. There is no general joint, polygon or angular-dynamics solver.

File → Settings → Input maps move_up/down/left/right, action, pause and back to remote buttons. Prefer action names in game code so mapping changes do not require rewriting gameplay. The simulator remote controls runtime input; the editing shortcuts in chapter 5 control authoring instead.

15. Audio authoring and playback

Audio imports/manages clips and buses. Select a clip to set volume, pitch, pan, loop, one-shot and play-on-awake. Mixer routes apply gain, mute/solo, EQ and supported effects. Mute silences a route; Solo isolates according to mixer rules. Preview auditions authoring sound, not device decoder behavior.

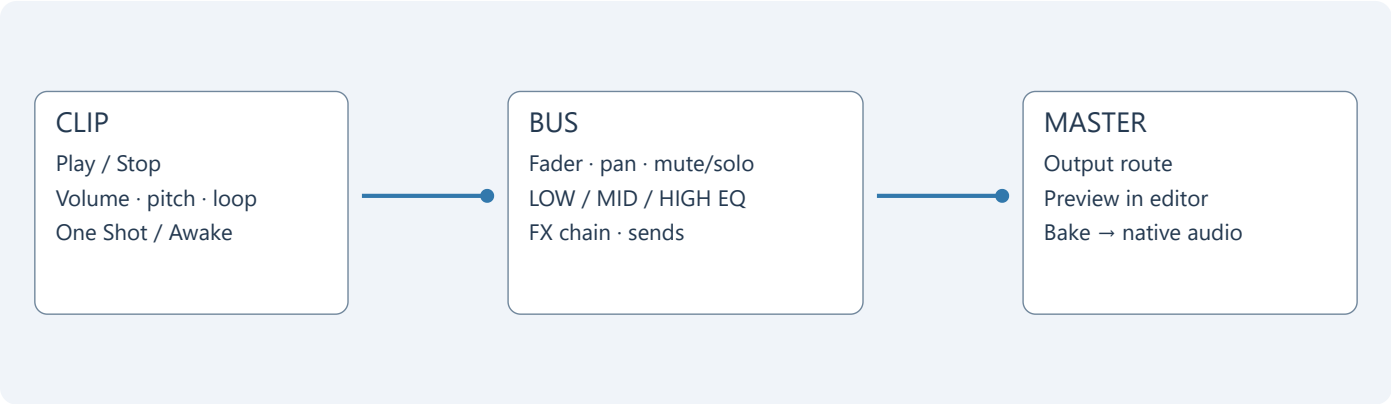


Figure 11. Illustration: clips route through buses to the master. Faders/knobs change sound; meters indicate activity rather than changing it.

Mixer controls and seven effects

Clear Solo clears solo isolation across buses. + Add Bus creates a submixer. A bus name can be renamed; its delete control removes a non-master bus. A fader changes gain; meters display signal activity. Pan positions sound left/right. Send routing selects an output bus and its send amount. Avoid routing cycles. EQ opens LOW (80 Hz), MID (1 kHz) and HIGH (12 kHz) gain sliders. FX opens the effect list; Add inserts the selected effect; the checkbox bypasses/enables it; the pencil opens its parameters; the cross removes it. + Import Clip imports project audio, while a clip row's transport auditions it. Parameter changes are saved as authoring metadata and baked during export.

Effect	Parameter meanings and defaults
Reverb	roomSize 0.5: apparent space; damping 0.5: damping of reverberation; wet 0.3: processed amount.
Delay	time 0.25 seconds: echo delay; feedback 0.4: repeated echo amount; wet 0.3: processed amount.
Compressor	threshold -24 dB: onset; ratio 4: compression ratio; attack 0.003 seconds and release 0.25 seconds: response timing.
Distortion	drive 0.5: saturation intensity; tone 0.5: tonal setting; wet 0.7: processed amount.
Chorus	rate 1.5 Hz: modulation speed; depth 0.7: modulation extent; feedback 0.3: recirculation; wet 0.5.

Effect	Parameter meanings and defaults
Phaser	rate 0.5 Hz: modulation speed; depth 0.6: sweep extent; feedback 0.7: recirculation; wet 0.5.
BitCrusher	bits 8: quantization resolution; normFreq 0.1: normalized sample-rate setting; wet 0.5.

Wet values run from 0 to 1. The panel's sliders show the supported range/units for each parameter. More effect intensity is not necessarily louder-safe audio: audition levels and verify the baked result. Effects are editor/export processing, not live device automation.

Export bakes mixer/effect processing into audio/native playback variants, preserving original files. It is not a live multi-voice DSP graph on Roku; changing baked parameters needs a rebuild. Loops retain their period after warm-up; non-loops can include effect tails. All seven supported editor FX are baked by the export audio path.

Use One Shot for short effects; compressed sources convert to WAV. Device low-latency slots determine overlap. Music uses one shared stream, so a second music stream replaces the first. Streaming supports play/pause/resume/stop; one-shots support play/stop. Test codec support, latency and memory on hardware.

Attach AudioSourceComponent and assign Resource Path. Play On Input/Trigger Action can activate it; Play On Awake comes from clip settings. Scripts can control it:

```
context.engine.ControlAudio("MusicNode", "play")
context.engine.ControlAudio("MusicNode", "pause")
context.engine.ControlAudio("MusicNode", "resume")
context.engine.ControlAudio("MusicNode", "stop")
```

An optional third argument selects a component ID for nodes with multiple sources. False indicates missing/disabled sources or unsupported commands. A small compressed source can expand greatly after decoding.

16. BrightScript editor and hooks

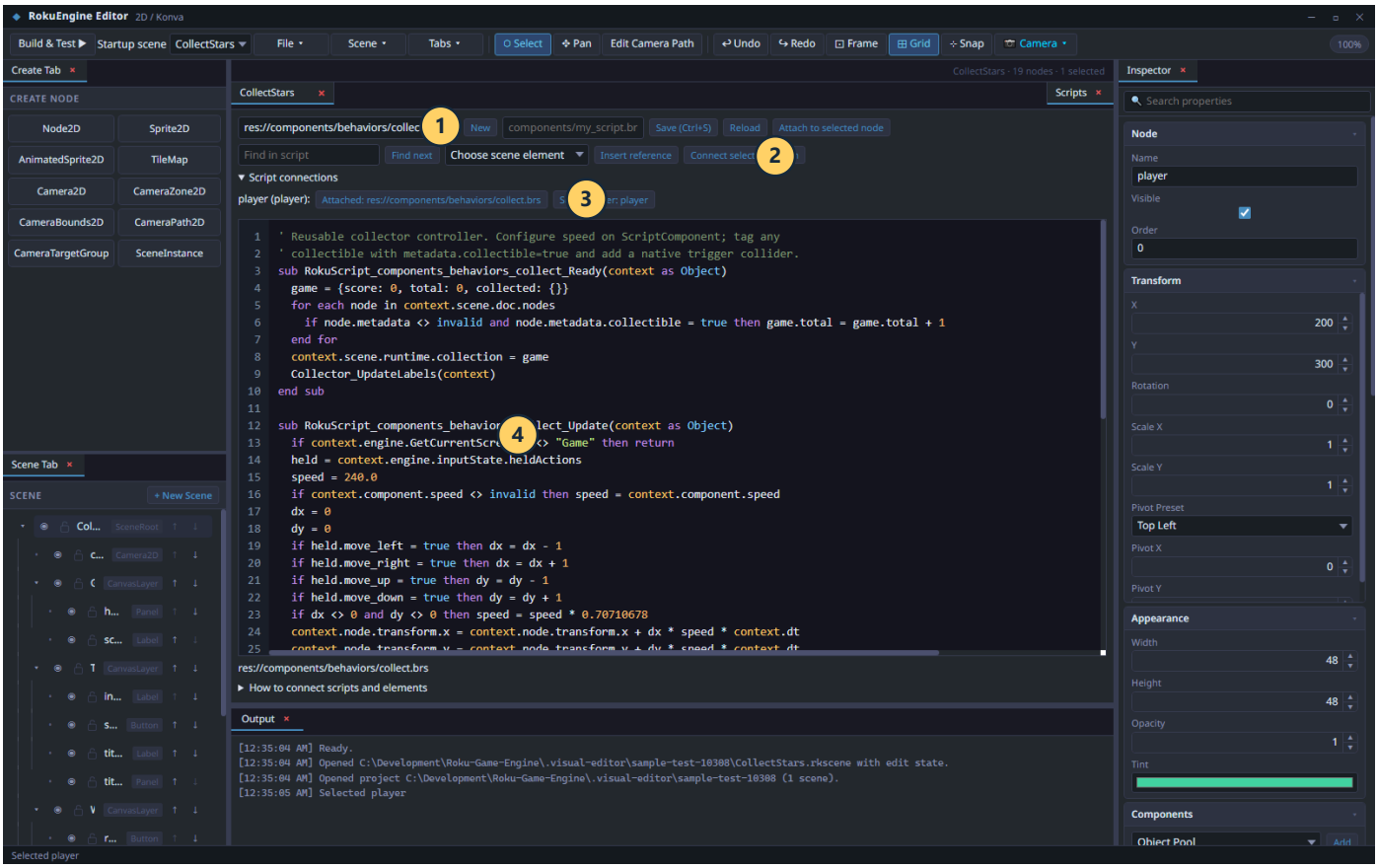


Figure 12. Scripts screenshot (Dark theme): 1 source/draft controls; 2 reference/button connection tools; 3 script owners/connections; 4 native BrightScript source.

Open Tabs → Scripts, double-click a project script in FileSystem, or choose Edit script on its component. New creates a .brs file under components. Select a node, then Attach to selected node to add ScriptComponent. Related/controller links describe architecture; they do not attach another executable script automatically.

Hook names use the relative path: components/player.brs becomes RokuScript_components_player. Update prefixes and component paths if moving/renaming a script. Generated source/main.brs and source/engine are outside the project script editor's normal editable source list.

```
sub RokuScript_components_player_Ready(context as Object)
  context.node.properties.opacity = 1.0
end sub

sub RokuScript_components_player_Update(context as Object)
  context.node.transform.x = context.node.transform.x + 60 * context.dt
end sub
```

Context contains node (owner), component, scene, engine, dt (seconds) and extra (event data). Each attachment has an owner context. Do not attach a central controller repeatedly unless it should run for each owner. Runtime mutations affect the running scene copy, not the saved authored template.

Source editing controls

The file dropdown switches drafts; Refresh updates the file list; Save writes source; Reload reads from disk and can replace a draft. Insert reference inserts a lookup for the selected element. Connect selected button attaches the script, replaces the button action and creates an Action hook. Owner buttons select owners; connection links open related source/hook positions. Find/Find next search. Compiler diagnostics appear at build time: syntax coloring is not compilation.

External changes block unsafe source overwrites. Copy important unsaved work before Reload. Saving an attachment requires saving scene data; saving a hook requires saving source. Build & Test saves edited scripts through this workflow before compiling.

```
sub RokuScript_components_player_Ready(context as Object)
  label = context.scene.nodeMap["ScoreLabel"]
  if label <> invalid then label.properties.text = "Score: 0"
end sub
```

Use the stable ID inserted by the reference tool and check for invalid. FindNodeByName is convenient but duplicate names are ambiguous.

Hook	Purpose and extra data
Ready / Init	Owner initialization; Init is a legacy alternative. Avoid duplicate initialization.
SceneEnter / SceneExit	Scene lifecycle and exit cleanup.
PreUpdate / PhysicsUpdate / Update / LateUpdate / PostUpdate	Corresponding variable/fixed-step phases; dt is seconds.
Input / Action	Remote events/actions; inspect extra and the intended owner.
Collision / CollisionEnter / CollisionStay / CollisionExit	Contacts; extras include otherNodeId/phase where supplied.
Timeout	Timer fired; waitTime.
PathFinished	Non-looping follower reached an endpoint; progress.
ScreenEntered / ScreenExited	Visibility region state, including initial state; targetNodeId.
Spawned	Successful spawn/acquisition; spawnedNodeId.
MenuSelectionChanged / MenuItemSelected	MenuList selection change/acceptance.

Hook	Purpose and extra data
AnimationEvent	Graph event; extra.event holds authored event data.
AnimationStateChanged	Destination stateId/clipId after a graph transition.
CameraZoneChanged	Zone notification when emitted by the camera event path.

17. Build, export, simulator and devices

F5, Export Game and device deployment share native export/validation. Current snapshots overlay unsaved scenes while unopened scenes remain available for transitions. References must exist inside the project and be package-safe; prefer res:// to machine-specific absolute paths.

Packages contain root manifest, source/main.brs, source/engine, scene JSON, resource metadata and assets. Generated builds live under .visual-editor/engine-builds. Edit project/canonical source instead of a generated copy that the next build will replace.

Simulator provides runtime output, remote controls, restart/stop and downloadable ZIP. Compiler failures must be fixed before interpreting missing gameplay as rendering errors. Check visibility, layout, resource paths and native warnings for invisible controls.

Deploy dock

Open Deploy, Add an IP or Scan, select Target and enter developer username/password. Deploy builds the current native project and uploads it. Debug/Dev emit runtime telemetry; Release disables it without stripping user print statements. Status, warnings, package size and history report the result.

Connect logs opens the native console; Disconnect stops it and automatic reconnection. Screenshot uses the selected device credentials. Restart returns Home and launches the installed developer channel; it does not uninstall it. The separate Sideload dialog uses the same native build/upload service.

The device/network must permit developer services. If scanning finds nothing, add the IP manually and check reachability. Connection, build and upload success are separate outcomes. Local transport tests are not physical-device or network certification.

18. Logs, memory, streaming and performance

Output records editing/build activity. Device Logs displays native console lines, filter/level choices, clear-display controls and available telemetry. Clearing a display does not clear device storage. A connected console is not proof of healthy frame rate.

Memory Estimates reads editor asset metadata and may use defaults when dimensions/duration are unavailable. Live telemetry reports cache/allocation estimates and frame timing when enabled. Neither measures total device RAM. Use these views to identify expensive content, then profile hardware.

Streaming processes a lazy queue in PreUpdate. Manifest preloadGroups.boot.textures queues at startup. Other groups can be requested by script. batchSize caps assets processed per frame, but one decode remains synchronous. budgetKb controls LRU bitmap eviction. Active scene textures/libraries/tilesets/styles remain pinned; lastWarning explains why a pinned set can exceed budget. Audio preload resolves metadata rather than inventing allocated memory.

```
context.engine.assetStream.PreloadGroup("level2", true)
```

Transformed sprite cache limits are 16 MiB total, 4 MiB per sprite and 2048 pixels/axis. UI clipping uses an extra full-screen surface. Repeated parallax, large textures, many animations and script work add cost. Resize artwork appropriately, reuse assets, pool frequent spawns and test the most crowded scenes.

19. Settings and appearance

File → Settings provides Project Name, Input, Screen, Physics and Appearance. Project Name is saved to project.settings.json. Input defines action mappings/profile. Screen selects mode, dimensions and fullscreen where supported. Physics chooses the default AABB/circle trigger contact method. Save writes project settings. Close/Cancel without Save does not commit that project-edit operation.

Appearance offers Dark, Light and Mid-tone. Mid-tone is the default when no valid preference exists; existing saved choices remain. Changes apply immediately to open editor windows and save automatically across projects, independently of project Save. They do not recolor artwork, game UI or node guides. A preference-save failure rolls back the change and reports the error.

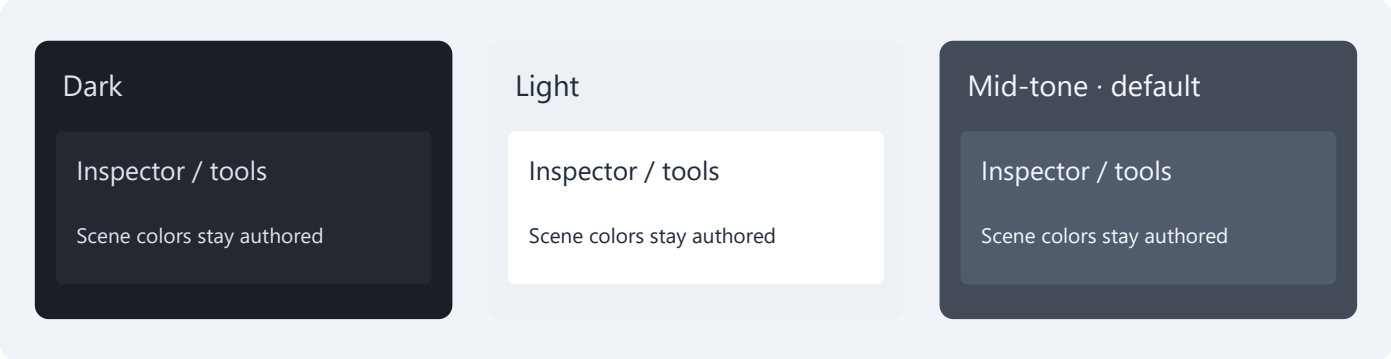


Figure 13. Palette illustration using the actual theme colors. Saved choices override the new Mid-tone default.

20. Worked example: title screen to game

1. Create/open a project and a Game scene. Add a player Sprite2D with a small image.
2. Attach an input component for simple movement or a script for continuous action-driven movement.
3. Add Collider2D and only the rigid-body behavior you need. Add static walls/ground.
4. Create Camera2D, enable Current and assign the player as its follow target. Test with F5.
5. Add CanvasLayer for a HUD. Create ScoreLabel and anchor it near the top-left.
6. Build a title screen/layer with Panel, title Label and Start Button. Use FocusGroup and Initial Focus.
7. Assign the matching screen/scene action or connect a controller script. scene:Game requires an actual scene named Game.
8. Add Timer/Spawner if needed; set Max Active and use pools for frequent disposable objects.
9. Configure Audio clips and AudioSource components; reserve the stream for music and one-shots for short effects.
10. Save scenes/source/resources, choose the startup scene, build/test, then verify the game on a real Roku.

Implement and test one system at a time: static visibility, movement, collision, camera, then UI. This separates resource-reference problems from input/physics problems.

21. Troubleshooting and recovery

Symptom	Checks/recovery
Editor fails to start	Launch with npm.cmd start; verify dependencies and local Konva.
Dock disappeared	Use Tabs or Tabs → Default.
Cannot select an item	Check editing lock, active workspace/tool and hierarchy; use the tree.
Child jumps during layout	Check anchors/margins, parent transforms and container-owned layout.
Blank artwork	Check path/file, visibility/opacity, dimensions, camera and build errors.
Frozen animation	Check library/clip, playing, speed scale and component/graph configuration.
Camera will not follow	Check Current/Enabled, target, bounds and zone/path overrides.
Menu cannot activate	Check disabled/focusable, group state, focus and input action map.
UI action does nothing	Match target scene/screen/option and script prefix/attachment.
Wrong tiles change	Check map, layer, brush, mode and tileset.
Script will not save	Check external edits/path; copy drafts before Reload.
Audio FX differ	Rebuild baked audio and check stream/one-shot limits on hardware.
Deployment fails	Check IP/network, developer credentials/services and compiler/package errors.
No live statistics	Use Debug/Dev telemetry and Connect logs; Release omits engine telemetry.
Cache budget exceeded	Reduce pinned assets/dimensions; estimates are not total RAM.
Runtime edit vanished on reload	Runtime state is temporary; save an authored edit for permanent content changes.

Recovery/autosave supports the project workflow but does not replace deliberate saves/backups. Preserve source assets, .rkscene files and editor-state sidecars together. Keep source-control history or copies before large inheritance/resource edits.

22. Native limits and release checklist

Camera view rotation, rotated UI and affine shear rendering are not implemented. Solid physics uses AABBs; trigger contacts and projectile queries can use boxes or circles. Audio FX are baked; streams/one-shots have device limits. Old WebGPU/ECS scaffolding is not the active editor/runtime. Do not assume an unused source module is an available feature.

Before release, verify startup scene, references, compilation, remote focus, Back/Pause, screen transitions, persistence, sound loops, long sessions, busy scenes and memory on intended hardware. Simulator success does not establish certification or device performance.

The following catalogs cover every creatable node/component and its schema-exposed settings. The control index covers static buttons/selects in the application pages; dynamic controls are covered by workflow chapters and catalogs. The script API chapter documents game-facing operations. Its final signature index is for advanced source navigation; internal method presence does not make it a stable game API.

Sprite Slicer live frame preview

Import an image or use the selected element image. The whole image is selected initially and the resized preview appears below the source. Use Grid or Draw rectangle to choose an atlas region. Set All slice margins for equal borders, then adjust Left, Top, Right and Bottom individually or drag the pink guides over the source image. Smaller margins produce thinner preserved borders. Output width and height resize the center while keeping the corner dimensions. Save selected regions after changing margins, then Apply or Create; the output dimensions are applied to the element. Use Panel or Button for nine-slice frames; Sprite2D and TextureRect use ordinary image stretching and accept regions without borders. Select a saved region to restore its margins.

Building a reliable scene-to-scene flow

Start with `samples/scene-to-scene` for a minimal Title ? Play ? Title example. Open both scene files to expose their names in the Action picker. Select the button and choose **Fade to scene: Play** or **Open scene: Play**. The UI designer offers matching Action/Back Action suggestions while allowing custom text. Set the toolbar **Startup scene** to the scene that should run first, then use F5 / Build & Test. Editing a scene does not by itself change the startup scene.

New CanvasLayers use a top-left origin at X=0/Y=0, so UI child positions are screen-relative. Explicit pivots saved in existing scenes are preserved. A layout container controls its visible UI children; position children freely under a Control or CanvasLayer when automatic layout is not desired. Hidden children do not occupy HBox/VBox slots. Nested containers resolve parents before children, and centered pivots retain the same visible rectangle as the runtime slot. UI rotation remains subject to the runtime limitations described in this manual.

Build diagnostics check scene names, UI screen names, MenuList item actions, and built-in Game/Options scene destinations. A successful build does not replace testing your actual button actions. The scene-workflow simulator check verifies default focus, transitions in both directions, and authored button placement; validate your final game on Roku hardware too.

Visual close-ups

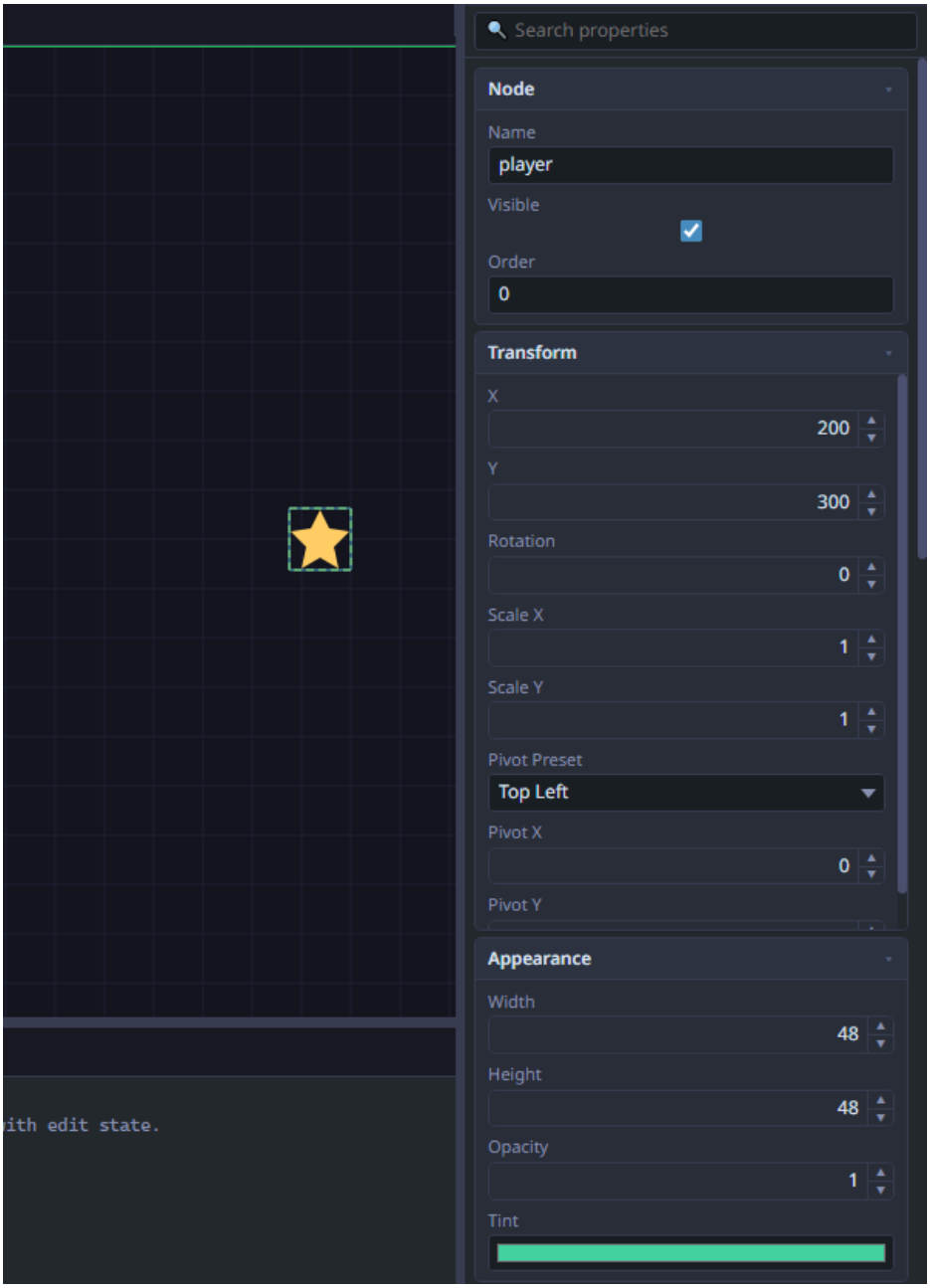


Figure 14. Enlarged Inspector screenshot. Search, section disclosures, text fields, checkboxes, numeric steppers and resource/color controls have different input roles.

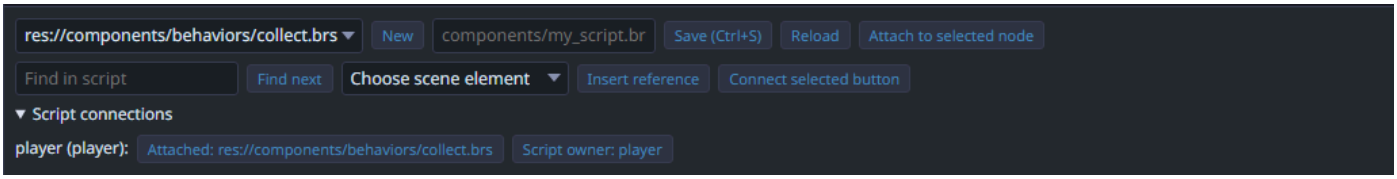


Figure 15. Enlarged Scripts toolbar: source file, New, Save, Reload, Attach; Find, scene reference, Insert reference and Connect selected button.

23. Node catalog — all creatable types

Property paths below are relative to node.properties. Common name/visibility/order/transform fields are explained in chapter 5. Unset means the schema does not supply an explicit value; optional resource and target fields must be configured when used. Some nodes also expose specialized array/resource editors (points, targets, items and layers) described in their workflow chapters.

Node2D

Core · `Node2D`

General world object and parent for related objects. It can provide a colored rectangle/group and a transform. Use components to give it behavior.

Field / default	Meaning and choices
Width `width` 160	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.
Height `height` 120	Rectangle height in local pixels. Minimum: 1.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Tint `tint` "#5e7594"	Authored artwork or helper-guide color; editor themes do not replace this value.

Sprite2D

Visual · `Sprite2D`

Static world image. Assign imagePath and set its rectangle/fit mode; transforms support rotation, signed scale, tint and inherited opacity.

Field / default	Meaning and choices
Width `width` 96	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.
Height `height` 96	Rectangle height in local pixels. Minimum: 1.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Image Path `imagePath` ""	Project-relative image resource. Prefer res://images/name.png; referenced files must exist inside the project.
Fit Mode `fitMode` "contain"	Contain preserves the whole image; cover fills and crops; stretch fills with possible aspect distortion. Choices: contain, cover, stretch.
Pick Image `imageDataUrl` ""	Embedded image preview/source data used by legacy imports; normally managed by image/resource tools.

AnimatedSprite2D

Visual · `AnimatedSprite2D`

Frame-based world artwork. Assign a SpriteFrames library and clip name. Use AnimationPlayerComponent for playback/graph controls.

Field / default	Meaning and choices
Width `width` 96	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.
Height `height` 96	Rectangle height in local pixels. Minimum: 1.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
SpriteFrames `spriteFramesId` ""	ID of an imported SpriteFrames library. The library supplies named clips and sheet references.
Animation `animation` ""	Clip name selected from the assigned SpriteFrames library; it must match existing metadata.
Frame `frame` 0	Manually selected frame/index when using non-playing or explicit-frame animation. Minimum: 0.
Playing `playing` true	Advance this animation/path while enabled. Turning this off pauses that playback.
Autoplay `autoplay` false	Initial automatic animation playback setting.
Speed Scale `speedScale` 1	Playback-rate multiplier. 1 is normal speed; 0 stops advancement. Minimum: 0.1.

TileMap

Visual · `TileMap`

Layered grid of tiles from a tileset. Author cells with the Tile Maps tool and edit tile rules/collision in the Tilesets and Tile Opts tools.

Field / default	Meaning and choices

Assigned TileSet
`tilesetId`
`""`
	ID of the tileset supplying tile images/rules.

Tile Width `tileWidth` 32	Width of one map/sheet cell in pixels. Minimum: 1.
Tile Height `tileHeight` 32	Height of one map/sheet cell in pixels. Minimum: 1.
Show Grid `showGrid` true	Authoring grid visibility; it is not extra tile artwork.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.

Camera2D

Core · `Camera2D`

Runtime world camera. Enable Current/Enabled and set a follow target or a static position. Editor guide dimensions are not the screen resolution.

Field / default	Meaning and choices
Current `current` false	Choose this camera as the active runtime camera.
Enabled `enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
Follow Target Name `followTargetName` ""	Node name or ID to track. Use an existing, unambiguous target.
Follow Lerp `followLerp` 1	Camera follow response parameter, normally 0–1. Larger values respond more directly. Minimum: 0. Maximum: 1.
Offset X `offsetX` 0	Horizontal offset in the relevant local/camera context.
Offset Y `offsetY` 0	Vertical offset in the relevant local/camera context.
Deadzone Width `deadzoneWidth` 0	Horizontal size of the target region that does not force camera movement. Minimum: 0.
Deadzone Height `deadzoneHeight` 0	Vertical size of the target region that does not force camera movement. Minimum: 0.
Deadzone Offset X `deadzoneOffsetX` 0	Horizontal shift of the camera deadzone.
Deadzone Offset Y `deadzoneOffsetY` 0	Vertical shift of the camera deadzone.

Field / default	Meaning and choices
Zoom `zoom` 1	Runtime camera magnification; distinct from authoring viewport zoom. Minimum: 0.1. Maximum: 10.
Camera Preset `cameraPreset` "Side Scroller"	Preset camera configuration. Review resulting individual settings for the intended game. Choices: Side Scroller, Metroidvania, Top Down Adventure, Twin Stick Shooter, RPG/JRPG, RTS, Fighting Game, Shmup, Pixel Perfect, Cinematic, Multi Target.
Look Ahead X `lookAheadX` 64	Horizontal anticipatory framing based on target movement direction.
Look Ahead Y `lookAheadY` 0	Vertical anticipatory framing based on target movement direction.
Smooth Follow `smoothFollow` true	Use camera follow smoothing rather than an abrupt change.
Use Camera Bounds `cameraBounds` true	Apply the camera bounds constraint.
Bounds Left `boundsX` Unset / optional	Left/world X of camera constraint.
Bounds Top `boundsY` Unset / optional	Top/world Y of camera constraint.
Bounds Width `boundsWidth` Unset / optional	Horizontal bounds/extent used by the relevant camera or AI behavior.
Bounds Height `boundsHeight` Unset / optional	Vertical bounds/extent used by the relevant camera or AI behavior.
Dynamic Zoom `dynamicZoom` false	Adjust zoom through the native dynamic camera behavior.

Field / default	Meaning and choices
Enable Screen Shake `screenShake` true	Permit camera shake requested through the runtime API.
Min Zoom `zoomMin` 0.5	Lower limit for dynamic camera zoom.
Max Zoom `zoomMax` 2	Upper limit for dynamic camera zoom.
Room Lock `roomLock` false	Use room-lock camera framing.
Combat Zoom `combatZoom` 1.2	Zoom used when the scene combat state selects combat framing.

CameraZone2D

Camera · `CameraZone2D`

Region that changes camera framing/behavior while the tracked target is inside. Priority selects between overlapping enabled zones.

Field / default	Meaning and choices
Width `width` 512	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.
Height `height` 288	Rectangle height in local pixels. Minimum: 1.
Enabled `enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
Mode `mode` "Normal"	Selected behavior mode for this node/tool. Native support depends on the documented system. Choices: Normal, Room Lock, Zoom Zone, Cinematic, Boss Arena.
Priority `priority` 0	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
Blend Time `blendTime` 0.5	Time in seconds for camera-zone changes to blend. Minimum: 0.
Zoom Override `zoom` 1	Runtime camera magnification; distinct from authoring viewport zoom. Minimum: 0.1. Maximum: 10.
Offset X `offsetX` 0	Horizontal offset in the relevant local/camera context.
Offset Y `offsetY` 0	Vertical offset in the relevant local/camera context.
Events Enabled `eventsEnabled` true	Permit notifications through the camera-zone event path.

Field / default	Meaning and choices
Show Debug `showDebug` true	Show available authoring/debug visualization; not ordinary game artwork.
Show Gizmo `showGizmo` true	Show the authoring helper/gizmo.

CameraBounds2D

Camera · `CameraBounds2D`

World-space region for limiting camera travel/zoom. Check the active camera bounds configuration and native behavior.

Field / default	Meaning and choices
Width `width` 1024	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.
Height `height` 768	Rectangle height in local pixels. Minimum: 1.
Clamp Position `clampPosition` true	Limit camera position to the configured bounds.
Clamp Zoom `clampZoom` true	Limit camera zoom through the bounds behavior.

CameraPath2D

Camera · `CameraPath2D`

Authored cinematic camera route. Edit points, interpolation and playback. Speed takes precedence over duration; speed below 1 enables duration-based playback.

Field / default	Meaning and choices
Closed Path `closedPath` false	Add a closing segment from the last point back to the first.
Loop Playback `loop` false	Repeat playback after reaching the end.
Camera Speed `speed` 200	Path units per second. Camera paths use duration when speed is below 1; gameplay followers support negative speed. Minimum: 1.
Interpolation `interpolation` "Linear"	Camera-route interpolation mode. Gameplay Path2D routes use straight segments. Choices: Linear, Smooth, Cubic.
Duration (s) `duration` 2	Camera-path fallback travel time in seconds; a configured speed normally takes precedence. Minimum: 0.1.
Auto Play `autoPlay` false	Initial automatic camera-path playback setting.

CameraTargetGroup

Camera · `CameraTargetGroup`

Target collection used to frame several objects. Assign the target list through the group editor; padding, smoothing and zoom limits shape the view.

Field / default	Meaning and choices
Auto Track `autoTrack` true	Enable automatic tracking of group targets.
Smoothing `smoothing` 0.15	Group-framing response/smoothing parameter. Minimum: 0. Maximum: 1.
Enable Auto Zoom `enableAutoZoom` true	Automatically fit the target group within zoom limits.
Min Zoom `minZoom` 0.5	Lower target-group zoom limit. Minimum: 0.1. Maximum: 10.
Max Zoom `maxZoom` 3	Upper target-group zoom limit. Minimum: 0.1. Maximum: 10.
Padding Left `paddingLeft` 64	Left inner/group-framing padding in pixels. Minimum: 0.
Padding Right `paddingRight` 64	Right inner/group-framing padding in pixels. Minimum: 0.
Padding Top `paddingTop` 64	Top inner/group-framing padding in pixels. Minimum: 0.
Padding Bottom `paddingBottom` 64	Bottom inner/group-framing padding in pixels. Minimum: 0.
Show Debug `showDebug` true	Show available authoring/debug visualization; not ordinary game artwork.

CanvasLayer

UI · `CanvasLayer`

Screen-space parent for HUDs and menus, independent of world camera travel. Layer controls UI order; uiScreenName associates it with a screen.

Field / default	Meaning and choices
Width `width` 1280	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.
Height `height` 720	Rectangle height in local pixels. Minimum: 1.
Layer `layer` 0	CanvasLayer: UI ordering level. Collision helpers/components: interaction category used with Mask.
Follow Viewport `followViewport` true	CanvasLayer viewport-follow authoring setting; UI descendants use screen-space runtime coordinates.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
UI Screen Name `uiScreenName` ""	Name used to associate the layer with a UI screen.

Control

UI · `Control`

General UI rectangle/parent with anchors, focus and optional clipping. It does not imply a button action or a texture by itself.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 240	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 64	Rectangle height in local pixels. Minimum: 1.
Focus Mode `focusMode` "none"	Whether/how this UI control participates in focus. Choices: none, click, all.
Clip Content `clipContent` false	Clip descendants to this control; nested clips intersect.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Focusable `focusable` false	Allow this control to be considered for remote focus.
Default Focus `defaultFocus` false	Prefer this control for initial focus when applicable.
Focus Up (id/name) `focusUp` ""	Explicit target for upward navigation (node reference).
Focus Down (id/name) `focusDown` ""	Explicit target for downward navigation.
Focus Left (id/name) `focusLeft` ""	Explicit target for left navigation.
Focus Right (id/name) `focusRight` ""	Explicit target for right navigation.

HBoxContainer

UI · `HBoxContainer`

Horizontal automatic layout of visible UI children. Separation and alignment control spacing; oversized children shrink to available width.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 320	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 64	Rectangle height in local pixels. Minimum: 1.
Separation `separation` 4	Gap in pixels between box-container children. Minimum: 0.
Alignment `alignment` "begin"	Begin/center/end alignment within available box extent. Choices: begin, center, end.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.

VBoxContainer

UI · `VBoxContainer`

Vertical automatic layout of visible UI children. Separation and alignment control spacing; oversized children shrink to available height.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 120	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 320	Rectangle height in local pixels. Minimum: 1.
Separation `separation` 4	Gap in pixels between box-container children. Minimum: 0.
Alignment `alignment` "begin"	Begin/center/end alignment within available box extent. Choices: begin, center, end.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.

MarginContainer

UI · `MarginContainer`

Padded layout parent. Descendants fill the inner rectangle after subtracting four padding values.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 280	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 180	Rectangle height in local pixels. Minimum: 1.
Padding Left `paddingLeft` 8	Left inner/group-framing padding in pixels. Minimum: 0.
Padding Top `paddingTop` 8	Top inner/group-framing padding in pixels. Minimum: 0.
Padding Right `paddingRight` 8	Right inner/group-framing padding in pixels. Minimum: 0.
Padding Bottom `paddingBottom` 8	Bottom inner/group-framing padding in pixels. Minimum: 0.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.

ScrollContainer

UI · `ScrollContainer`

Clipped UI viewport. Runtime scrollOffsetX/Y choose the visible part of content. Authoring scrollbar modes do not imply a native mouse scrollbar.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 320	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 240	Rectangle height in local pixels. Minimum: 1.
Horizontal Scroll `scrollHorizontal` "auto"	Authoring horizontal-scroll policy. Runtime content uses scrollOffsetX. Choices: auto, always, never.
Vertical Scroll `scrollVertical` "auto"	Authoring vertical-scroll policy. Runtime content uses scrollOffsetY. Choices: auto, always, never.
Scroll Offset X `scrollOffsetX` 0	Horizontal content offset in pixels; positive values expose content further to the right.
Scroll Offset Y `scrollOffsetY` 0	Vertical content offset in pixels; positive values expose content further down.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.

Label

UI · `Label`

Text display for titles, instructions, HUD values and data bindings. Set rectangle, typography, colors and alignment; it is not automatically focusable.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 240	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 36	Rectangle height in local pixels. Minimum: 1.
Text `text` "Label"	Displayed text. Runtime scripts can update properties.text.
Font Size `fontSize` 24	Requested text size in pixels; provide sufficient control height. Minimum: 8.
Text Color `textColor` "#e7eef7"	Text RGBA/RGB color as accepted by the editor.
Align `align` "left"	Horizontal text alignment. Choices: left, center, right.
Vertical Align `valign` "center"	Vertical text alignment. Choices: top, center, bottom.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Bound Option Key `boundOptionKey` Unset / optional	Option key displayed by this bound label/control; the option definition must exist.
Option Label Prefix `boundOptionPrefix` Unset / optional	Text placed before the bound option display.
High Score Table ID `boundHighScoreTable` Unset / optional	Existing local high-score table to display.
High Score Rank `boundHighScoreRank` Unset / optional	Rank/row selected from the bound high-score table. Minimum: 1.
Font family `fontFamily` Unset / optional	Requested font-family name for authoring/native font selection where supported.

Field / default	Meaning and choices
Font file (res://fonts/...) `fontPath` Unset / optional	Project font path, normally res://fonts/name.ttf.
Bold `fontBold` Unset / optional	Request bold text where supported by the chosen font.
Italic `fontItalic` Unset / optional	Request italic text where supported by the chosen font.

Panel

UI · `Panel`

Colored or nine-slice-styled UI rectangle. Useful for backgrounds/cards and grouping; ordinary Panel has no built-in button activation.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 280	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 120	Rectangle height in local pixels. Minimum: 1.
Color `color` "#263247"	Surface/background color or texture modulation according to node type.
Border Color `borderColor` "#5e7594"	Outline color for the control.
Border Width `borderWidth` 2	Outline thickness in local pixels. Minimum: 0.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Nine-Slice Style `nineSliceStyleId` Unset / optional	ID of a reusable UI atlas/nine-slice style.

TextureRect

UI · `TextureRect`

UI image with contain/cover/stretch fitting. Cropped reusable atlas regions can be applied without replacing existing transforms.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 160	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 90	Rectangle height in local pixels. Minimum: 1.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Image Path `imagePath` ""	Project-relative image resource. Prefer res://images/name.png; referenced files must exist inside the project.
Fit Mode `fitMode` "contain"	Contain preserves the whole image; cover fills and crops; stretch fills with possible aspect distortion. Choices: contain, cover, stretch.
Pick Image `imageDataUrl` ""	Embedded image preview/source data used by legacy imports; normally managed by image/resource tools.

Button

UI · `Button`

Rectangular remote-focusable action control with normal, focused, pressed and disabled states. Set an action or connect an attached script.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 220	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 56	Rectangle height in local pixels. Minimum: 1.
Text `text` "Button"	Displayed text. Runtime scripts can update properties.text.
Text Color `textColor` "#ffffff"	Text RGBA/RGB color as accepted by the editor.
Normal Color `normalColor` "#375a7f"	Button/menu surface when not focused/selected.
Focused Color `focusedColor` "#5b8def"	Button surface while it owns focus.
Pressed Color `pressedColor` "#8ab4ff"	Button surface during activation.
Disabled Color `disabledColor` "#4b5563"	Surface used for a disabled button.
Disabled `disabled` false	Prevent normal activation and use the disabled state.
Focus Mode `focusMode` "all"	Whether/how this UI control participates in focus. Choices: none, click, all.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Nine-Slice Style `nineSliceStyleId` Unset / optional	ID of a reusable UI atlas/nine-slice style.
Focusable `focusable` true	Allow this control to be considered for remote focus.

Field / default	Meaning and choices
Default Focus `defaultFocus` false	Prefer this control for initial focus when applicable.
Action `action` ""	Action string dispatched on activation; see the action table in chapter 11.
Back Action `actionBack` ""	Action associated with the Back operation where handled by the UI system.
Focus Up (id/name) `focusUp` ""	Explicit target for upward navigation (node reference).
Focus Down (id/name) `focusDown` ""	Explicit target for downward navigation.
Focus Left (id/name) `focusLeft` ""	Explicit target for left navigation.
Focus Right (id/name) `focusRight` ""	Explicit target for right navigation.
Cycles Option Key `boundOptionKey` Unset / optional	Option key displayed by this bound label/control; the option definition must exist.
Font size `fontSize` Unset / optional	Requested text size in pixels; provide sufficient control height. Minimum: 1.
Font family `fontFamily` Unset / optional	Requested font-family name for authoring/native font selection where supported.
Font file (res://fonts/...) `fontPath` Unset / optional	Project font path, normally res://fonts/name.ttf.
Bold `fontBold` Unset / optional	Request bold text where supported by the chosen font.

Field / default	Meaning and choices
Italic `fontItalic` Unset / optional	Request italic text where supported by the chosen font.
Text alignment `align` Unset / optional	Horizontal text alignment. Choices: left, center, right.
Vertical alignment `valign` Unset / optional	Vertical text alignment. Choices: top, center, bottom.

CircleButton

UI · `CircleButton`

Circular/elliptical button artwork using the normal button action/focus system. Keep equal dimensions for a circle.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Diameter (W) `width` 80	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Diameter (H) `height` 80	Rectangle height in local pixels. Minimum: 1.
Text `text` ""	Displayed text. Runtime scripts can update properties.text.
Text Color `textColor` "#ffffff"	Text RGBA/RGB color as accepted by the editor.
Normal Color `normalColor` "#375a7f"	Button/menu surface when not focused/selected.
Focused Color `focusedColor` "#5b8def"	Button surface while it owns focus.
Pressed Color `pressedColor` "#8ab4ff"	Button surface during activation.
Disabled Color `disabledColor` "#4b5563"	Surface used for a disabled button.
Disabled `disabled` false	Prevent normal activation and use the disabled state.
Focus Mode `focusMode` "all"	Whether/how this UI control participates in focus. Choices: none, click, all.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Focusable `focusable` true	Allow this control to be considered for remote focus.
Default Focus `defaultFocus` false	Prefer this control for initial focus when applicable.

Field / default	Meaning and choices
Action `action` ""	Action string dispatched on activation; see the action table in chapter 11.
Back Action `actionBack` ""	Action associated with the Back operation where handled by the UI system.
Focus Up (id/name) `focusUp` ""	Explicit target for upward navigation (node reference).
Focus Down (id/name) `focusDown` ""	Explicit target for downward navigation.
Focus Left (id/name) `focusLeft` ""	Explicit target for left navigation.
Focus Right (id/name) `focusRight` ""	Explicit target for right navigation.
Cycles Option Key `boundOptionKey` Unset / optional	Option key displayed by this bound label/control; the option definition must exist.
Font size `fontSize` Unset / optional	Requested text size in pixels; provide sufficient control height. Minimum: 1.
Font family `fontFamily` Unset / optional	Requested font-family name for authoring/native font selection where supported.
Font file (res://fonts/...) `fontPath` Unset / optional	Project font path, normally res://fonts/name.ttf.
Bold `fontBold` Unset / optional	Request bold text where supported by the chosen font.
Italic `fontItalic` Unset / optional	Request italic text where supported by the chosen font.

Field / default	Meaning and choices
Text alignment `align` Unset / optional	Horizontal text alignment. Choices: left, center, right.
Vertical alignment `valign` Unset / optional	Vertical text alignment. Choices: top, center, bottom.

TriangleButton

UI · `TriangleButton`

Triangular button artwork using the normal button action/focus system. Useful for play/directional visual motifs.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 80	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 80	Rectangle height in local pixels. Minimum: 1.
Text `text` ""	Displayed text. Runtime scripts can update properties.text.
Text Color `textColor` "#ffffff"	Text RGBA/RGB color as accepted by the editor.
Normal Color `normalColor` "#375a7f"	Button/menu surface when not focused/selected.
Focused Color `focusedColor` "#5b8def"	Button surface while it owns focus.
Pressed Color `pressedColor` "#8ab4ff"	Button surface during activation.
Disabled Color `disabledColor` "#4b5563"	Surface used for a disabled button.
Disabled `disabled` false	Prevent normal activation and use the disabled state.
Focus Mode `focusMode` "all"	Whether/how this UI control participates in focus. Choices: none, click, all.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Focusable `focusable` true	Allow this control to be considered for remote focus.
Default Focus `defaultFocus` false	Prefer this control for initial focus when applicable.

Field / default	Meaning and choices

Action
`action`
`""`
	Action string dispatched on activation; see the action table in chapter 11.

Back Action
`actionBack`
`""`
	Action associated with the Back operation where handled by the UI system.

Focus Up (id/name)
`focusUp`
`""`
	Explicit target for upward navigation (node reference).

Focus Down (id/name)
`focusDown`
`""`
	Explicit target for downward navigation.

Focus Left (id/name)
`focusLeft`
`""`
	Explicit target for left navigation.

Focus Right (id/name)
`focusRight`
`""`
	Explicit target for right navigation.

Cycles Option Key `boundOptionKey` Unset / optional	Option key displayed by this bound label/control; the option definition must exist.
Font size `fontSize` Unset / optional	Requested text size in pixels; provide sufficient control height. Minimum: 1.
Font family `fontFamily` Unset / optional	Requested font-family name for authoring/native font selection where supported.
Font file (res://fonts/...) `fontPath` Unset / optional	Project font path, normally res://fonts/name.ttf.
Bold `fontBold` Unset / optional	Request bold text where supported by the chosen font.
Italic `fontItalic` Unset / optional	Request italic text where supported by the chosen font.

Field / default	Meaning and choices
Text alignment `align` Unset / optional	Horizontal text alignment. Choices: left, center, right.
Vertical alignment `valign` Unset / optional	Vertical text alignment. Choices: top, center, bottom.

OvalButton

UI · `OvalButton`

Elliptical button artwork using the normal button action/focus system. Text/state styling follows the other buttons.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 160	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 56	Rectangle height in local pixels. Minimum: 1.
Text `text` "Button"	Displayed text. Runtime scripts can update properties.text.
Text Color `textColor` "#ffffff"	Text RGBA/RGB color as accepted by the editor.
Normal Color `normalColor` "#375a7f"	Button/menu surface when not focused/selected.
Focused Color `focusedColor` "#5b8def"	Button surface while it owns focus.
Pressed Color `pressedColor` "#8ab4ff"	Button surface during activation.
Disabled Color `disabledColor` "#4b5563"	Surface used for a disabled button.
Disabled `disabled` false	Prevent normal activation and use the disabled state.
Focus Mode `focusMode` "all"	Whether/how this UI control participates in focus. Choices: none, click, all.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Focusable `focusable` true	Allow this control to be considered for remote focus.
Default Focus `defaultFocus` false	Prefer this control for initial focus when applicable.

Field / default	Meaning and choices

Action
`action`
`""`
	Action string dispatched on activation; see the action table in chapter 11.

Back Action
`actionBack`
`""`
	Action associated with the Back operation where handled by the UI system.

Focus Up (id/name)
`focusUp`
`""`
	Explicit target for upward navigation (node reference).

Focus Down (id/name)
`focusDown`
`""`
	Explicit target for downward navigation.

Focus Left (id/name)
`focusLeft`
`""`
	Explicit target for left navigation.

Focus Right (id/name)
`focusRight`
`""`
	Explicit target for right navigation.

Cycles Option Key `boundOptionKey` Unset / optional	Option key displayed by this bound label/control; the option definition must exist.
Font size `fontSize` Unset / optional	Requested text size in pixels; provide sufficient control height. Minimum: 1.
Font family `fontFamily` Unset / optional	Requested font-family name for authoring/native font selection where supported.
Font file (res://fonts/...) `fontPath` Unset / optional	Project font path, normally res://fonts/name.ttf.
Bold `fontBold` Unset / optional	Request bold text where supported by the chosen font.
Italic `fontItalic` Unset / optional	Request italic text where supported by the chosen font.

Field / default	Meaning and choices
Text alignment `align` Unset / optional	Horizontal text alignment. Choices: left, center, right.
Vertical alignment `valign` Unset / optional	Vertical text alignment. Choices: top, center, bottom.

NineSlicePanel

UI · `NineSlicePanel`

Textured surface whose corners/edges are preserved when resizing. Slice values must leave a usable center inside the source rectangle.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 320	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 200	Rectangle height in local pixels. Minimum: 1.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Tint Color `color` "#ffffff"	Surface/background color or texture modulation according to node type.
Image Path `imagePath` ""	Project-relative image resource. Prefer res://images/name.png; referenced files must exist inside the project.
Pick Image `imageDataUrl` ""	Embedded image preview/source data used by legacy imports; normally managed by image/resource tools.
Slice Left `sliceLeft` 16	Preserved left border inside the source texture/region. Minimum: 0.
Slice Top `sliceTop` 16	Preserved top border. Minimum: 0.
Slice Right `sliceRight` 16	Preserved right border. Minimum: 0.
Slice Bottom `sliceBottom` 16	Preserved bottom border. Minimum: 0.
Center Mode `centerMode` Unset / optional	How the central nine-slice area fills. Choices: stretch, tile.
Edge Mode `edgeMode` Unset / optional	How the nine-slice edge strips fill. Choices: stretch, tile.
Nine-Slice Style `nineSliceStyleId` Unset / optional	ID of a reusable UI atlas/nine-slice style.

MenuList

UI · `MenuList`

Scrollable visible window of menu rows with selectedIndex. Built-in up/down skips disabled entries, OK uses item-specific/fallback action.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 280	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 240	Rectangle height in local pixels. Minimum: 1.
Item Height `itemHeight` 48	Height of each MenuList row in pixels. Minimum: 16.
Selected Index `selectedIndex` 0	Zero-based selected MenuList entry. Minimum: 0.
Focus Mode `focusMode` "all"	Whether/how this UI control participates in focus. Choices: none, click, all.
Item Color `normalColor` "#263247"	Button/menu surface when not focused/selected.
Selected Color `selectedColor` "#375a7f"	Surface color of the selected row.
Text Color `textColor` "#e7eef7"	Text RGBA/RGB color as accepted by the editor.
Selected Text Color `selectedTextColor` "#ffffff"	Text color of the selected row.
Font Size `fontSize` 22	Requested text size in pixels; provide sufficient control height. Minimum: 8.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Focusable `focusable` true	Allow this control to be considered for remote focus.
Default Focus `defaultFocus` false	Prefer this control for initial focus when applicable.

Field / default	Meaning and choices

Action (on select)
`action`
`""`
	Action string dispatched on activation; see the action table in chapter 11.

Focus Up (id/name)
`focusUp`
`""`
	Explicit target for upward navigation (node reference).

Focus Down (id/name)
`focusDown`
`""`
	Explicit target for downward navigation.

Focus Left (id/name)
`focusLeft`
`""`
	Explicit target for left navigation.

Focus Right (id/name)
`focusRight`
`""`
	Explicit target for right navigation.

ProgressBar

UI · `ProgressBar`

Value display between minValue and maxValue, optionally labeled/skinned. Set its value from scripts; it is not an interactive slider by default.

Field / default	Meaning and choices
Anchor Preset `anchorPreset` Unset / optional	Convenience selection of anchor positions/stretch behavior.
Anchor Left `anchorLeft` 0	Left anchor as fraction of parent width (0 left, 1 right). Minimum: 0. Maximum: 1.
Anchor Top `anchorTop` 0	Top anchor as fraction of parent height (0 top, 1 bottom). Minimum: 0. Maximum: 1.
Anchor Right `anchorRight` 0	Right anchor as fraction of parent width. Minimum: 0. Maximum: 1.
Anchor Bottom `anchorBottom` 0	Bottom anchor as fraction of parent height. Minimum: 0. Maximum: 1.
Margin Left `marginLeft` 0	Pixel offset from the left anchor.
Margin Top `marginTop` 0	Pixel offset from the top anchor.
Margin Right `marginRight` 0	Pixel offset from the right anchor.
Margin Bottom `marginBottom` 0	Pixel offset from the bottom anchor.
Width `width` 300	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.

Field / default	Meaning and choices
Height `height` 28	Rectangle height in local pixels. Minimum: 1.
Value `value` 0.5	Current progress value, interpreted between Min Value and Max Value. Minimum: 0. Maximum: 1.
Min Value `minValue` 0	Lower progress range bound.
Max Value `maxValue` 1	Upper progress range bound.
Show Label `showLabel` false	Show the progress text label.
Fill Color `fillColor` "#4a90d9"	Filled portion color.
Track Color `trackColor` "#1a2638"	Unfilled/background track color.
Border Color `borderColor` "#5e7594"	Outline color for the control.
Border Width `borderWidth` 1	Outline thickness in local pixels. Minimum: 0.
Label Color `labelColor` "#e7eef7"	Progress label color.
Font Size `fontSize` 18	Requested text size in pixels; provide sufficient control height. Minimum: 8.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.

SceneInstance

Core · `SceneInstance`

Reusable scene reference with instance overrides. Edit the source for shared changes or override a particular instance; repair missing references.

Field / default	Meaning and choices
Width `width` 160	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type. Minimum: 1.
Height `height` 120	Rectangle height in local pixels. Minimum: 1.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity. Minimum: 0. Maximum: 1.
Source Scene `sourceScenePath` ""	Project-relative reusable scene resource.
Source Name `sourceSceneName` ""	Name of the referenced scene.
Editable Children `editableChildren` true	Permit instance-child editing/overrides through the editor workflow.

Marker2D

Gameplay · `Marker2D`

Named local/world point for spawns/checkpoints/attachments. Its crosshair/label are editing guides rather than game artwork.

Field / default	Meaning and choices
Gizmo Size `gizmoSize` 16	Marker crosshair size in the authoring view.
Tint `tint` "#f2cf66"	Authored artwork or helper-guide color; editor themes do not replace this value.

Timer

Gameplay · `Timer`

Logical countdown/repeating timer. Attach Timeout hooks, use seconds, and choose scene-pause behavior. Scripts can restart or stop it.

Field / default	Meaning and choices
Wait Time `waitTime` 1	Timer interval/countdown in seconds; must be positive.
One Shot `oneShot` false	Stop this timer after one firing (audio clips use an independent One Shot setting).
Autostart `autostart` true	Begin this timer/spawner automatically.
Paused `paused` false	Pause this timer without removing it.
Process When Paused `processWhenPaused` false	Advance the timer using frame time even while the scene is paused.
Tint `tint` "#70c9d1"	Authored artwork or helper-guide color; editor themes do not replace this value.

TriggerZone2D

Gameplay · `TriggerZone2D`

Trigger region with selectable project-default, box or circle contacts. Use layer/mask to choose interactions and collision hooks to respond.

Field / default	Meaning and choices
Width `width` 160	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type.
Height `height` 120	Rectangle height in local pixels.
Enabled `enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
Contact Method `contactMethod` "project"	For trigger contacts and queries: project inherits Settings > Physics; aabb uses box bounds; circle uses distance checks. Solid-body movement remains AABB. Choices: project, aabb, circle.
Circle Radius `circleRadius` 0	Local circle radius. Zero fits half the smaller Width/Height. The center uses rectangle midpoint plus offsets; world radius uses the largest transform stretch.
Layer `layer` "default"	CanvasLayer: UI ordering level. Collision helpers/components: interaction category used with Mask.
Mask `mask` "default"	Collision category filter paired with Layer.
Tint `tint` "#6dbd79"	Authored artwork or helper-guide color; editor themes do not replace this value.

Path2D

Gameplay · `Path2D`

Gameplay route made of local straight-line segments. Parent transforms move/rotate/scale the route; edit points through Inspector/Edit Path.

Field / default	Meaning and choices
Points `points` [{"x":0,"y":0},{"x":160,"y":0}]	Array of local {x,y} waypoints; edit through the point tools/Inspector.
Closed Path `closedPath` false	Add a closing segment from the last point back to the first.
Tint `tint` "#66c9e8"	Authored artwork or helper-guide color; editor themes do not replace this value.

PathFollow2D

Gameplay · `PathFollow2D`

Moves along an assigned/parent Path2D. Parent moving artwork beneath it; progress is path-local distance and can be changed to seek.

Field / default	Meaning and choices

Path (node ID or name; blank = parent)
`path`
`""`
	ID/name of Path2D; empty can use the Path2D parent.

Speed `speed` 60	Path units per second. Camera paths use duration when speed is below 1; gameplay followers support negative speed.
Progress `progress` 0	Distance along the gameplay route in path-local units; changing it seeks.
Loop `loop` true	Repeat playback after reaching the end.
Rotates `rotates` false	Rotate the follower to align with its route tangent.
Playing `playing` true	Advance this animation/path while enabled. Turning this off pauses that playback.
Tint `tint` "#66c9e8"	Authored artwork or helper-guide color; editor themes do not replace this value.

Parallax2D

World · `Parallax2D`

Camera-relative background parent with optional autoscroll/repeat. Repeats descendant artwork rather than duplicating collisions.

Field / default	Meaning and choices
Width `width` 320	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type.
Height `height` 180	Rectangle height in local pixels.
Scroll Scale X `scrollScaleX` 0.5	Horizontal camera-relative motion: 0 screen-fixed, 1 normal world motion.
Scroll Scale Y `scrollScaleY` 0.5	Vertical camera-relative motion: 0 screen-fixed, 1 normal world motion.
Autoscroll X `autoscrollX` 0	Additional horizontal world motion per second.
Autoscroll Y `autoscrollY` 0	Additional vertical world motion per second.
Horizontal repeat size (0 = off) `repeatWidth` 0	Local horizontal repeat spacing; 0 disables this axis.
Vertical repeat size (0 = off) `repeatHeight` 0	Local vertical repeat spacing; 0 disables this axis.
Tint `tint` "#b496ff"	Authored artwork or helper-guide color; editor themes do not replace this value.

VisibilityEnabler2D

Gameplay · `VisibilityEnabler2D`

Region that suspends a target subtree offscreen. Margin extends the active view; multiple regions combine to keep a shared target awake.

Field / default	Meaning and choices
Width `width` 160	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type.
Height `height` 120	Rectangle height in local pixels.
Target (node ID or name; blank = parent) `target` ""	Target node ID/name; an empty visibility target can use the parent.
Enabled `enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
Margin `margin` 64	Extra camera-region margin used before suspending the visibility target.
Tint `tint` "#72bdab"	Authored artwork or helper-guide color; editor themes do not replace this value.

Spawner2D

Gameplay · `Spawner2D`

Clones a template hierarchy or acquires from a named pool at marker/origin positions. Limit active instances and choose automatic/manual bursts.

Field / default	Meaning and choices

Template node (ID or name)
`template`
`""`
	Existing template node ID/name whose subtree is cloned.

Pool name (optional)
`poolName`
`""`
	Name of a prepared ObjectPool pool; use instead of template cloning.

Spawn markers (node IDs or names)
`spawnPoints`
`[]`
	Ordered marker references cycled between spawn requests; empty uses spawner origin.

Interval `interval` 1	Seconds between automatic spawn bursts.
Max Active `maxActive` 10	Maximum number of live spawned/acquired roots.
Spawn Count `spawnCount` 1	Number of requests in each burst.
Autostart `autostart` true	Begin this timer/spawner automatically.
Enabled `enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.

Tint
`tint`
`"#f0a266"`
	Authored artwork or helper-guide color; editor themes do not replace this value.

FocusGroup

UI · `FocusGroup`

Menu navigation scope. It can trap/wrap focus and restore previous selection when reopened; disabled groups remove descendant focus targets.

Field / default	Meaning and choices
Width `width` 320	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type.
Height `height` 240	Rectangle height in local pixels.
Opacity `opacity` 1	Opacity multiplier from 0 (transparent) to 1 (opaque), combined with inherited opacity.
Enabled `enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
Initial focus (node ID or name) `initialFocus` ""	Initial descendant control reference within this FocusGroup.
Wrap `wrap` true	Wrap focus/menu selection at the ends.
Restore Focus `restoreFocus` true	Restore the last valid focus when reopening the group.
Trap Focus `trapFocus` true	Keep navigation inside this group by default.
Tint `tint` "#e5c97a"	Authored artwork or helper-guide color; editor themes do not replace this value.

24. Component catalog — all creatable behaviors

These values live on a component object, not inside node.properties. Common Enabled/Priority apply within the component phase. IDs identify a particular component when a node owns more than one. Defaults are authoring defaults, not a guarantee that every combination has useful gameplay behavior.

Object Pool

`ObjectPoolComponent`

Pre-spawns reusable node instances. Call engine.PoolAcquire(name,x,y) and engine.PoolRelease(nodeId) from scripts.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 10	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`pools` "[{\"name\":\"Bullet\",\"nodeType\":\"Sprite2D\",\"size\":30},{\"name\":\"Explosion\",\"nodeType\":\"AnimatedSprite2D\",\"size\":10}]"	JSON pool definitions with name, nodeType and preallocated size. Exhaustion returns invalid rather than allocating unlimited objects.

Enemy AI

`EnemyAIComponent`

Drives enemy movement with a selectable AI pattern.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 90	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`aiType` "patrol"	Enemy movement pattern selected for this component; inspect its relevant pattern parameters.
`speed` 60	Path units per second. Camera paths use duration when speed is below 1; gameplay followers support negative speed.
`targetName` "Player"	Existing target node name/ID.
`patrolWidth` 160	Patrol movement extent.
`wanderMoveTime` 1.8	Seconds spent moving during a wander cycle.
`wanderWaitTime` 0.8	Seconds spent waiting during a wander cycle.
`alertRadius` 120	Distance at which the movement behavior reacts to its target.
`leashRadius` 200	Distance limiting pursuit before returning.
`returnSpeed` 40	Movement speed while returning.
`stopDistance` 8	Distance to target at which approach movement stops.
`turnSpeed` 2.5	Pattern turn-rate parameter.
`boundsWidth` 1920	Horizontal bounds/extent used by the relevant camera or AI behavior.

Setting / default	Meaning
`boundsHeight` 1080	Vertical bounds/extent used by the relevant camera or AI behavior.
`orbitRadius` 80	Radius for orbit movement.
`sineAmplitude` 40	Displacement of sine-pattern motion.
`sineFrequency` 2	Rate of sine-pattern oscillation.
`moveAxisX` 0	Horizontal movement axis/direction component.
`moveAxisY` -1	Vertical movement axis/direction component.
`flankDistance` 80	Offset distance used by flank-pattern movement.

Move

`MoveComponent`

Moves a node every frame using velocity.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 120	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`vx` 0	Horizontal velocity in units per second.
`vy` 0	Vertical velocity in units per second.

Spin

`SpinComponent`

Rotates a node continuously.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 130	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`degreesPerSecond` 90	Continuous rotation rate in degrees/second.

Bob

`BobComponent`

Offsets a node with a sine wave.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 140	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`axis` "y"	Axis receiving the sine/bob offset.
`amplitude` 16	Sine/bob displacement size.
`speed` 2	Path units per second. Camera paths use duration when speed is below 1; gameplay followers support negative speed.
`phase` 0	Initial sine/bob phase.

Follow Input

`FollowInputComponent`

Moves a node when Roku remote button events fire.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 100	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`speed` 24	Path units per second. Camera paths use duration when speed is below 1; gameplay followers support negative speed.
`controlsCamera` false	Direct movement input toward the camera path when supported by this component.

Collider2D

`Collider2DComponent`

Adds box solid physics and selectable box or circle trigger contacts.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 150	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`width` 32	Rectangle width in local pixels; interpreted as a guide/content rectangle according to node type.
`height` 32	Rectangle height in local pixels.
`offsetX` 0	Horizontal offset in the relevant local/camera context.
`offsetY` 0	Vertical offset in the relevant local/camera context.
`isTrigger` false	Report overlaps instead of treating this collider as a solid.
`contactMethod` "project"	For trigger contacts and queries: project inherits Settings > Physics; aabb uses box bounds; circle uses distance checks. Solid-body movement remains AABB.
`circleRadius` 0	Local circle radius. Zero fits half the smaller Width/Height. The center uses rectangle midpoint plus offsets; world radius uses the largest transform stretch.
`layer` "default"	CanvasLayer: UI ordering level. Collision helpers/components: interaction category used with Mask.
`mask` "default"	Collision category filter paired with Layer.
`debug` false	Enable available collider debug visualization.

RigidBody2D

`RigidBody2DComponent`

Applies simple velocity, acceleration, gravity, and collision resolution.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 120	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`bodyType` "dynamic"	Dynamic/static/kinematic body behavior.
`vx` 0	Horizontal velocity in units per second.
`vy` 0	Vertical velocity in units per second.
`ax` 0	Horizontal acceleration in units per second squared.
`ay` 0	Vertical acceleration in units per second squared.
`gravity` 0	Downward acceleration supplied by the rigid-body behavior.
`friction` 1	Velocity damping/friction parameter used by the native solver.
`bounce` 0	Restitution/bounce parameter.
`mass` 1	Relative body mass used in collision velocity response.
`lockRotation` true	Prevent rigid-body rotation through the Spin component path.

Script

`ScriptComponent`

Runs BrightScript hooks. Use Edit script to open its source.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 300	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`scriptPath` ""	Project-relative .brs script path; hooks must use the prefix derived from this path.
`entryPoint` "Update"	Configured script entry hook; lifecycle/event hooks also follow the dispatch rules.
`exposeNode` true	Script-component owner-exposure setting used by the script context workflow.

AudioSource

`AudioSourceComponent`

References an audio clip from the Audio Mixer. Playback settings (volume, pitch, loop, bus, etc.) are configured in the Audio tab.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 400	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`resourcePath` ""	Audio resource path selected from project/audio clip metadata.
`playOnInput` false	Play this AudioSource when Trigger Action is pressed.
`triggerAction` "action"	Mapped action name that triggers this component behavior.

InputController

`InputControllerComponent`

Maps Roku remote button presses to node movement.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 110	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`moveStep` 24	Input-controller movement amount.
`controlsCamera` false	Direct movement input toward the camera path when supported by this component.
`allowHorizontal` true	Permit horizontal mapped movement.
`allowVertical` true	Permit vertical mapped movement.

AnimationPlayer

`AnimationPlayerComponent`

Controls AnimatedSprite2D playback state and animation selection.

Setting / default	Meaning
`enabled` true	Enable this helper/behavior; disabled nodes may stop contributing to their system.
`priority` 350	Processing/selection priority. For overlapping camera zones the highest enabled priority wins.
`animation` ""	Clip name selected from the assigned SpriteFrames library; it must match existing metadata.
`graphPath` ""	Project-relative animation graph JSON path; empty uses ordinary clip playback.
`graphParameters` "{}"	JSON object providing initial graph parameter values.
`autoplay` true	Initial automatic animation playback setting.
`playing` true	Advance this animation/path while enabled. Turning this off pauses that playback.
`speedScale` 1	Playback-rate multiplier. 1 is normal speed; 0 stops advancement.
`triggerAction` "action"	Mapped action name that triggers this component behavior.

25. Button, selector and icon control index

Each entry uses the visible label and, where available, its original tooltip. The code-sized identifier helps distinguish repeated labels such as Save, Fit or Clear. Dropdown choices are listed for static options; scene/resource lists populate from the current project. Numeric/text fields and dynamically generated properties are documented in the catalogs. Keyboard-only shortcuts are in chapter 5.

Main editor

Control / location	Purpose
— `btn-minimize` Toolbar / scene · chapters 4–5	Minimize this desktop window to the taskbar.
□ `btn-maximize` Toolbar / scene · chapters 4–5	Toggle between maximized and restored window size.
× `btn-close` Toolbar / scene · chapters 4–5	Close this window; preserve unsaved work before exiting.
Build & Test ▶ `btn-test-simulator` Toolbar / scene · chapters 4–5	Build native Roku game and test it (F5).
Startup scene `startup-scene-select` Editor workspace	Choose the runtime entry scene for the next native build.
File ▾ `btn-file-menu` Menus / docking · chapters 3–4	Open project, export, sideload and settings commands.
Create Project `menu-create-project` Menus / docking · chapters 3–4	Run Create Project through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Open Project `menu-open-project` Menus / docking · chapters 3–4	Run Open Project through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Back to Manager `menu-back-to-manager` Menus / docking · chapters 3–4	Run Back to Manager through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.

Control / location	Purpose
Save Project `menu-save-project` Menus / docking · chapters 3–4	Run Save Project through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Save As `menu-save-project-as` Menus / docking · chapters 3–4	Run Save As through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Export Game `menu-export-game` Menus / docking · chapters 3–4	Run Export Game through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Build & Test in Roku Simulator (F5) `menu-test-simulator` Menus / docking · chapters 3–4	Run Build & Test in Roku Simulator (F5) through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Sideload to Roku ▸ `menu-sideload` Menus / docking · chapters 3–4	Run Sideload to Roku ▸ through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Settings `menu-settings` Menus / docking · chapters 3–4	Run Settings through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Quit `menu-quit` Menus / docking · chapters 3–4	Run Quit through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Scene ▾ `btn-scene-menu` Menus / docking · chapters 3–4	Open scene creation, saving, inheritance and close commands.
New Scene `menu-new-scene` Menus / docking · chapters 3–4	Run New Scene through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
New Inherited Scene `menu-new-inherited-scene` Menus / docking · chapters 3–4	Run New Inherited Scene through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Rename Scene `menu-rename-scene` Menus / docking · chapters 3–4	Run Rename Scene through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Open Scene `menu-open-scene` Menus / docking · chapters 3–4	Run Open Scene through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.

Control / location	Purpose
Save Scene `menu-save-scene` Menus / docking · chapters 3–4	Run Save Scene through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Save Scene As `menu-save-scene-as` Menus / docking · chapters 3–4	Run Save Scene As through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Reload Saved Scene `menu-reload-saved-scene` Menus / docking · chapters 3–4	Run Reload Saved Scene through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Rebase Inherited Scene `menu-rebase-inherited-scene` Menus / docking · chapters 3–4	Run Rebase Inherited Scene through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Revert Inherited To Base `menu-revert-inherited-scene` Menus / docking · chapters 3–4	Run Revert Inherited To Base through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Close Scene `menu-close-scene` Menus / docking · chapters 3–4	Run Close Scene through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Close All Scenes `menu-close-all-scenes` Menus / docking · chapters 3–4	Run Close All Scenes through its menu workflow; see chapters 4–5 for save/reload/inheritance consequences.
Tabs ▾ `btn-tabs-menu` Menus / docking · chapters 3–4	Show/hide dockable tools and restore the default arrangement.
Default `tabs-menu-default` Menus / docking · chapters 3–4	Restore default docking/tab arrangement.
⬡ **Select** Editor workspace	Select and manipulate nodes in the world canvas.
✥ **Pan** Editor workspace	Pan the editing viewport without changing game node coordinates.
Edit Path `btn-edit-camera-path` Camera tools · chapter 9	Edit Path2D or CameraPath2D points.

Control / location	Purpose
↩ Undo `btn-undo` Toolbar / scene · chapters 4–5	Undo the most recent supported authoring edit in the active editor.
↪ Redo `btn-redo` Toolbar / scene · chapters 4–5	Reapply the most recent undone authoring edit.
⊡ Frame `btn-frame` Toolbar / scene · chapters 4–5	Frame the current selection in the world editing view; it does not move the selected node.
⊞ Grid `btn-grid` Toolbar / scene · chapters 4–5	Toggle the world editing grid.
⊹ Snap `btn-snap` Toolbar / scene · chapters 4–5	Toggle world editing snapping.
📷 Camera ▾ `btn-camera-overlays` Camera tools · chapter 9	Open camera guide-visibility and camera-view navigation commands.
⊡ Focus Camera `cam-tool-focus` Camera tools · chapter 9	Focus the world editing view on the camera.
⊞ Frame Camera `cam-tool-frame` Camera tools · chapter 9	Frame the camera view within the world editing viewport.
↗ Jump To Camera `cam-tool-jump` Camera tools · chapter 9	Move the editing view to the active camera position.
Toggle Preview Panel `btn-camera-preview` Camera tools · chapter 9	Show/hide the authoring camera preview panel.
Scene Tab Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
FileSystem Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Create Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.

Control / location	Purpose
Sprites Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Audio Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
+ New Scene `btn-add-child` Toolbar / scene · chapters 4–5	Create a new scene through the Scene tab shortcut (the visible label is + New Scene).
× `scene-node-search-clear` Editor workspace	Clear search.
Upload `btn-import-sheet` Sprites · chapter 6	Add or import the named item in this panel; configure its resource/selection first.
Grid `btn-sheet-grid` Sprites · chapter 6	Toggle the sprite-sheet cell grid overlay.
sheet-sort `sheet-sort` Sprites · chapter 6	Sort sprite sheets by name or cell count. *Choices:* Name A–Z; Name Z–A; Largest First; Smallest First
sheet-category-filter `sheet-category-filter` Sprites · chapter 6	Filter the sprite-sheet list by category. *Choices:* All Categories; Player; Enemies; Level; UI; Uncategorized
Main Scene Editor workspace	Switch the center workspace between world scene editing and UI screen editing.
UI Screens Editor workspace	Switch the center workspace between world scene editing and UI screen editing.
UI Screens Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Scripts Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Sprite Slicer Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Import image `slicer-pick` Sprite Slicer · chapter 6	Import a source image for slicing.

Control / location	Purpose
Use selected element image `slicer-use` Sprite Slicer · chapter 6	Use the selected image element as the slicer source.
slicer-mode `slicer-mode` Sprite Slicer · chapter 6	Choose regular grid slicing or manual rectangle creation. *Choices:* Grid; Draw rectangle
Slice grid `slicer-grid` Sprite Slicer · chapter 6	Generate complete cells using dimensions/spacing/offsets.
Select all `slicer-all` Sprite Slicer · chapter 6	Select all current slice regions.
Clear selection `slicer-none` Sprite Slicer · chapter 6	Clear slice selection without deleting image data.
Save selected regions `slicer-save` Sprite Slicer · chapter 6	Save named reusable regions; source artwork stays unchanged.
slicer-zoom `slicer-zoom` Sprite Slicer · chapter 6	Change slicer magnification without resampling source art. *Choices:* 25%; 50%; 100%; 200%; 400%
Apply region to selected element `slicer-apply` Sprite Slicer · chapter 6	Apply the selected saved region while preserving the element transform and existing connections.
Create element type `slicer-create-type` Sprite Slicer · chapter 6	Choose the node type created from a region. *Choices:* Sprite2D; TextureRect; Button; Panel
Create from region `slicer-create` Sprite Slicer · chapter 6	Make the chosen element type at the region natural size.
Edit UI borders / states `slicer-ui-editor` Sprite Slicer · chapter 6	Open detailed UI border/state editing for the style.
Project scripts `script-files` Scripts · chapter 16	Select an editable project BrightScript draft. *Choices:* Choose a script

Control / location	Purpose
New `script-new` Scripts · chapter 16	Create the .brs file at New script path and generate its hook prefix.
Save (Ctrl+S) `script-save` Scripts · chapter 16	Write the active script source; scene attachment saving is separate.
Reload `script-reload` Scripts · chapter 16	Read source from disk again. Copy unsaved edits first.
Attach to selected node `script-attach` Scripts · chapter 16	Add an executable ScriptComponent on the selected node.
Find next `script-find-next` Scripts · chapter 16	Move to the next match for the Find text.
Scene element reference `script-nodes` Scripts · chapter 16	Choose a scene element whose reference is inserted into the script.
Insert reference `script-reference` Scripts · chapter 16	Insert a stable-ID lookup for the chosen scene element at the text cursor.
Connect selected button `script-button` Scripts · chapter 16	Attach/connect the selected button to the active script and replace its existing action.
Expand editor `ui-expand-editor` UI Screens · chapters 10–11	Expand the UI Screens workspace; Return to dock restores its docked placement.
Screen to edit `ui-screen-select` UI Screens · chapters 10–11	Choose the Screen to edit setting/resource for this panel; choices are listed below.
Fit preview `ui-btn-refresh-canvas` UI Screens · chapters 10–11	Fit/recalculate the UI screen preview in its available area.
Delete selected `ui-btn-delete-node` UI Screens · chapters 10–11	Remove/clear the selected data in this panel. Check its current selection before using it.

Control / location	Purpose
⬡ **Layer** UI Screens · chapters 10–11	Add CanvasLayer.
◻ **Control** UI Screens · chapters 10–11	Add Control.
▭ **Panel** UI Screens · chapters 10–11	Add Panel.
T Label UI Screens · chapters 10–11	Add Label.
⊙ **Button** UI Screens · chapters 10–11	Add Button.
◯ **Circle** UI Screens · chapters 10–11	Add CircleButton.
△ **Triangle** UI Screens · chapters 10–11	Add TriangleButton.
⬭ **Oval** UI Screens · chapters 10–11	Add OvalButton.
⬚ **Texture** UI Screens · chapters 10–11	Add TextureRect.
⊞ **9Slice** UI Screens · chapters 10–11	Add NineSlicePanel.
▬ **Progress** UI Screens · chapters 10–11	Add ProgressBar.
≡ **MenuList** UI Screens · chapters 10–11	Add MenuList.
FocusGroup UI Screens · chapters 10–11	Add a remote navigation focus group.
↔ **HBox** UI Screens · chapters 10–11	Add HBoxContainer.
↕ **VBox** UI Screens · chapters 10–11	Add VBoxContainer.
⊡ **Margin** UI Screens · chapters 10–11	Add MarginContainer.

Control / location	Purpose
⇕ Scroll UI Screens · chapters 10–11	Add ScrollContainer.
Clear Guides `ui-clear-guides` UI Screens · chapters 10–11	Delete custom UI editing guides; game artwork is unchanged.
Preview resolution `ui-resolution-select` UI Screens · chapters 10–11	Choose an authoring preview resolution; verify actual project Screen settings separately. *Choices:* 1280×720 (720p); 1920×1080 (1080p); 960×540 (half); 640×360 (360p)
Fit `ui-zoom-fit` UI Screens · chapters 10–11	Change authoring preview magnification, leaving source assets and game camera zoom unchanged.
− `ui-zoom-out` UI Screens · chapters 10–11	Change authoring preview magnification, leaving source assets and game camera zoom unchanged.
+ `ui-zoom-in` UI Screens · chapters 10–11	Change authoring preview magnification, leaving source assets and game camera zoom unchanged.
Refresh `ui-screen-refresh` UI Screens · chapters 10–11	Refresh the UI authoring view.
Output Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Deploy Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Device Logs Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Memory Estimates Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Node Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
SpriteFrames Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Tile Sets Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.

Control / location	Purpose
Tile Maps Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
UI Assets Atlas Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Transitions Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Options & Scores Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Screen Templates Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Camera Preview Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
+ Add `sf-btn-add-anim` Editor workspace	Add animation.
✎ `sf-btn-rename-anim` Editor workspace	Rename selected animation.
✕ `sf-btn-del-anim` Editor workspace	Delete selected animation.
▶ `sf-btn-play` Editor workspace	Play preview.
■ `sf-btn-stop` Editor workspace	Stop preview.
sf-playback-mode `sf-playback-mode` Editor workspace	Choose the sf-playback-mode setting/resource for this panel; choices are listed below. *Choices:* Loop; Once; Ping-Pong
+ Frame `sf-btn-add-frames` Editor workspace	Add selected tile from Sprites panel as a frame.

Control / location	Purpose
− Frame `sf-btn-del-frame` Editor workspace	Remove selected frame.
Clear All `sf-btn-clear-frames` Editor workspace	Clear all frames from clip.
+ Create `ts-btn-create` Tiles/terrain/collision · chapter 8	Create the named tileset from the selected sheet/setup.
Rules `ts-overlay-rules` Tiles/terrain/collision · chapter 8	Show terrain rule indicators on tiles.
Collision `ts-overlay-collision` Tiles/terrain/collision · chapter 8	Show collision indicators on tiles.
Select w/ Rules `ts-select-rules-all` Tiles/terrain/collision · chapter 8	Select tiles that contain terrain rules.
Clear Selection `ts-select-clear` Tiles/terrain/collision · chapter 8	Clear tileset selection only.
tm-tileset-select `tm-tileset-select` Tiles/terrain/collision · chapter 8	Assign/select the tileset for map painting.
Sync Screen `tm-btn-resize` Tiles/terrain/collision · chapter 8	Synchronize TileMap dimensions with the current screen setup.
Clear `tm-btn-clear` Tiles/terrain/collision · chapter 8	Clear map data in the current editing context; inspect the active map/layer first.
+ Layer `tm-btn-layer-add` Tiles/terrain/collision · chapter 8	Add another TileMap layer.
Paint `tm-tool-paint` Tiles/terrain/collision · chapter 8	Select the Paint tile painting operation for the active map/layer.

Control / location	Purpose
Rect `tm-tool-rectangle` Tiles/terrain/collision · chapter 8	Select the Rect tile painting operation for the active map/layer.
Line `tm-tool-line` Tiles/terrain/collision · chapter 8	Select the Line tile painting operation for the active map/layer.
Erase `tm-tool-erase` Tiles/terrain/collision · chapter 8	Select the Erase tile painting operation for the active map/layer.
Fill `tm-tool-fill` Tiles/terrain/collision · chapter 8	Select the Fill tile painting operation for the active map/layer.
Pick `tm-tool-pick` Tiles/terrain/collision · chapter 8	Select the Pick tile painting operation for the active map/layer.
Tiles `tm-mode-tile` Tiles/terrain/collision · chapter 8	Select Tiles editing mode before painting.
Terrain `tm-mode-terrain` Tiles/terrain/collision · chapter 8	Select Terrain editing mode before painting.
Collision `tm-mode-collision` Tiles/terrain/collision · chapter 8	Select Collision editing mode before painting.
+ `nse-btn-add` UI Assets Atlas · chapter 12	Create a new style.
Pick Image `nse-btn-pick-image` UI Assets Atlas · chapter 12	Assign the style source texture.
nse-preview-mode `nse-preview-mode` UI Assets Atlas · chapter 12	Choose the style preview type. *Choices:* Original; Stretched; Button; Panel
nse-center-mode `nse-center-mode` UI Assets Atlas · chapter 12	Choose center-area fill behavior. *Choices:* Stretch; Tile

Control / location	Purpose
nse-edge-mode `nse-edge-mode` UI Assets Atlas · chapter 12	Choose edge-strip fill behavior. *Choices:* Stretch; Tile
Save Style `nse-btn-save` UI Assets Atlas · chapter 12	Persist the current reusable nine-slice/atlas style.
Delete `nse-btn-delete` UI Assets Atlas · chapter 12	Remove the selected style; references may need repair.
+ `te-btn-add` Transitions · chapter 12	Create a transition preset.
▶ Preview `te-btn-preview` Transitions · chapter 12	Audition the selected transition; this is not a scene build.
Duplicate `te-btn-dupe` Transitions · chapter 12	Duplicate the selected preset.
Delete `te-btn-delete` Transitions · chapter 12	Remove the selected transition preset.
te-type `te-type` Transitions · chapter 12	Choose the te-type setting/resource for this panel; choices are listed below. *Choices:* Fade; Slide Left; Slide Right; Slide Up; Slide Down; Wipe; Instant
te-easing `te-easing` Transitions · chapter 12	Choose the te-easing setting/resource for this panel; choices are listed below. *Choices:* Ease In-Out; Ease In; Ease Out; Linear; Elastic Out
⚙ Options `oe-tab-options` Options / High Scores · chapter 12	Show persistent option definitions.
🏆 High Scores `oe-tab-highscores` Options / High Scores · chapter 12	Show local high-score table definitions.
★ `oe-btn-add-builtin` Options / High Scores · chapter 12	Insert the built-in option definitions.

Control / location	Purpose
+ `oe-btn-add-opt` Options / High Scores · chapter 12	Create a custom persistent option definition.
× `oe-btn-del-opt` Options / High Scores · chapter 12	Delete the selected option definition.
+ `oe-btn-add-hs` Options / High Scores · chapter 12	Create a local high-score table definition.
× `oe-btn-del-hs` Options / High Scores · chapter 12	Delete the selected table definition.
✦ Create in Project `tpl-btn-create` UI Templates · chapter 12	Generate editable nodes from the selected UI template.
Select camera to preview `camera-preview-selector` Camera tools · chapter 9	Choose which camera the authoring preview displays.
⊡ `camera-preview-focus` Camera tools · chapter 9	Focus viewport on this camera.
⊞ `camera-preview-frame` Camera tools · chapter 9	Frame camera in viewport.
↺ `camera-preview-reset` Camera tools · chapter 9	Reset preview.
Play Path `camera-path-play` Camera tools · chapter 9	Preview playback of the selected camera path.
Stop `camera-path-stop` Camera tools · chapter 9	Stop the authoring path preview.
Fit `cpv-zoom-fit` Camera tools · chapter 9	Fit to panel.

Control / location	Purpose
100% `cpv-zoom-100` Camera tools · chapter 9	100%.
200% `cpv-zoom-200` Camera tools · chapter 9	200%.
400% `cpv-zoom-400` Camera tools · chapter 9	400%.
⏸ `cpv-pause` Camera tools · chapter 9	Pause/resume the camera authoring preview, not a deployed game.
↺ `cpv-refresh` Camera tools · chapter 9	Refresh preview.
Inspector Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
Tile Opts Dock tabs · chapter 3	Show/select this tool or settings tab. Opening a tool does not change the running game.
ts-sheet-select `ts-sheet-select` Tiles/terrain/collision · chapter 8	Select the sprite sheet used by the tileset.
Sync From Sheet `ts-btn-sync-sheet` Tiles/terrain/collision · chapter 8	Synchronize tileset metadata from the selected sprite sheet.
▭ **Rect** `tcs-tool-rect` Tiles/terrain/collision · chapter 8	Draw rectangle.
⬠ **Poly** `tcs-tool-polygon` Tiles/terrain/collision · chapter 8	Draw polygon (click points, double-click to close).
↖ **Select** `tcs-tool-select` Tiles/terrain/collision · chapter 8	Select / move shapes.

Control / location	Purpose
Full Tile `tcs-preset-full` Tiles/terrain/collision · chapter 8	Add full-tile collision box.
Clear All `tcs-clear-all` Tiles/terrain/collision · chapter 8	Remove/clear the selected data in this panel. Check its current selection before using it.
▭ **Rect** `tps-tool-rect` Tiles/terrain/collision · chapter 8	Draw rectangle.
⬠ **Poly** `tps-tool-polygon` Tiles/terrain/collision · chapter 8	Draw polygon (click points, double-click to close).
↖ **Select** `tps-tool-select` Tiles/terrain/collision · chapter 8	Select / move shapes.
Full Tile `tps-preset-full` Tiles/terrain/collision · chapter 8	Add full-tile physics box.
Clear All `tps-clear-all` Tiles/terrain/collision · chapter 8	Remove/clear the selected data in this panel. Check its current selection before using it.

settings

Control / location	Purpose
Project Name Settings · chapter 19	Show/select this tool or settings tab. Opening a tool does not change the running game.
Input Settings · chapter 19	Show/select this tool or settings tab. Opening a tool does not change the running game.
Screen Settings · chapter 19	Show/select this tool or settings tab. Opening a tool does not change the running game.
Physics Settings · chapter 19	Show/select this tool or settings tab. Opening a tool does not change the running game.
Appearance Settings · chapter 19	Show/select this tool or settings tab. Opening a tool does not change the running game.

Control / location	Purpose
contact-method `contact-method` Settings · chapter 19	Choose the contact-method setting/resource for this panel; choices are listed below. *Choices:* AABB (box bounds); Circle (distance checks)
editor-theme `editor-theme` Settings · chapter 19	Choose Dark, Light or default Mid-tone. This application preference saves immediately across projects. *Choices:* Dark; Light; Mid-tone
input-movement `input-movement` Settings · chapter 19	Choose the input-movement setting/resource for this panel; choices are listed below. *Choices:* Standard D-Pad; Platformer; Twin Stick Style
input-action `input-action` Settings · chapter 19	Choose the input-action setting/resource for this panel; choices are listed below. *Choices:* OK; Play/Pause; Options
input-pause `input-pause` Settings · chapter 19	Choose the input-pause setting/resource for this panel; choices are listed below. *Choices:* Play/Pause; Back; Options
action-move-up `action-move-up` Settings · chapter 19	Choose the action-move-up setting/resource for this panel; choices are listed below. *Choices:* Up; OK; Play/Pause; Options
action-move-down `action-move-down` Settings · chapter 19	Choose the action-move-down setting/resource for this panel; choices are listed below. *Choices:* Down; OK; Play/Pause; Options
action-move-left `action-move-left` Settings · chapter 19	Choose the action-move-left setting/resource for this panel; choices are listed below. *Choices:* Left; OK; Play/Pause; Options
action-move-right `action-move-right` Settings · chapter 19	Choose the action-move-right setting/resource for this panel; choices are listed below. *Choices:* Right; OK; Play/Pause; Options
action-back `action-back` Settings · chapter 19	Choose the action-back setting/resource for this panel; choices are listed below. *Choices:* Back; Options; Play/Pause
screen-mode `screen-mode` Settings · chapter 19	Choose the screen-mode setting/resource for this panel; choices are listed below. *Choices:* 720p; 1080p
Close `btn-cancel` Settings · chapter 19	Close the Settings page without saving its project fields; appearance has already saved automatically.

Control / location	Purpose
Save Settings `btn-save` Settings · chapter 19	Save the project settings on the Settings page.

sideload

Control / location	Purpose
Ping `btn-ping` Sideload · chapter 17	Check whether the configured device responds.
Save `btn-save-device` Sideload · chapter 17	Persist the current values through this panel save operation.
Cancel `btn-cancel-sideload` Sideload · chapter 17	Cancel/stop the current sideload workflow as offered by the dialog; check status for any operation already completed.
Close `btn-close` Sideload · chapter 17	Close this window; preserve unsaved work before exiting.
⬆ Sideload `btn-sideload` Sideload · chapter 17	Build/upload through the developer-device workflow using the selected target and credentials.
✕ `${d.id}` Sideload · chapter 17	Remove.

project manager

Control / location	Purpose
Settings `pm-settings-btn` Project manager · chapter 2	Open the project-manager settings workflow.
Close `pm-close-btn` Project manager · chapter 2	Close the project manager window.

Control / location	Purpose
Create `pm-create-btn` Project manager · chapter 2	Create a new project folder through the project-manager workflow.
Import `pm-import-btn` Project manager · chapter 2	Add an existing project folder to the manager.
Scan Folder Projects `pm-scan-btn` Project manager · chapter 2	Scan a selected local folder for projects, not the network for Roku devices.
pm-sort-select `pm-sort-select` Project manager · chapter 2	Choose the pm-sort-select setting/resource for this panel; choices are listed below. *Choices:* Last Edited; Name; Path
Edit Project manager · chapter 2	Open the selected project in the editor.
Export Project manager · chapter 2	Export the selected project through native build.
Rename Project manager · chapter 2	Rename the selected project folder through the manager workflow.
Duplicate Project manager · chapter 2	Duplicate the selected project folder.
Manage Tags Project manager · chapter 2	Edit comma-separated project tags.
Remove Project manager · chapter 2	Remove the project from the manager list; the project folder is not deleted.

Dynamic panel controls

Panel / controls	Purpose
Deploy: Target, username/password, profile	Select a device, authenticate the developer installer and choose telemetry behavior.
Deploy: Add / Scan / Deploy	Register an IP / discover devices / build and upload the current project.
Deploy: Connect logs / Disconnect / Screenshot / Restart	Open TCP logs / stop reconnection / capture the selected device / relaunch the installed developer channel.

Panel / controls	Purpose
Device Logs: Filter / All, Errors, Warnings, Info / Clear / Auto	Filter displayed lines, clear the view and choose automatic scrolling.
Memory Estimates: Refresh	Recalculate the current asset estimate. It does not sample total device RAM.
Audio: clip/bus selection, transport, volume, pan, pitch, loop, One Shot, mute/solo, FX/EQ	Author/audition sound, choose routes and configure processing baked during native export.
Scene row: disclosure / eye / lock / name / type	Expand children, change visibility/edit protection, identify and select the node.
Inspector: component selector / Add / enable checkbox / Edit script / remove	Attach/manage behavior components and open project source. Removal affects the selected owner.
Path points: add / point selection / move / delete	Edit the node-local route, independently of moving its parent transform.
UI properties: resource/font/color pickers, layout presets, alignment, snap/guides	Configure the selected element using the layout and styling concepts in chapters 10–12.
Saved slice/style/terrain/clip lists: select / add / rename / remove	Manage the selected reusable resource in its own editor context; save scene/resource changes afterward.

26. BrightScript API reference

Call these methods through context.engine. Signatures below are taken directly from the canonical runtime. BrightScript optional arguments include their defaults. Check node/resource results for invalid; use valid scene/screen/pool/option names. Stored changes are device data; normal node mutations are temporary runtime state.

Method signature	Purpose
`QueueSceneChange(mode as String, sceneName = invalid as Dynamic)`	Request a scene change at the safe frame boundary. Use a valid authored scene name.
`SetPaused(isPaused as Boolean)`	Pause/unpause scene processing according to phase rules.
`SetPhysicsPaused(isPaused as Boolean)`	Pause/unpause the fixed-step physics simulation.
`SetTimeScale(scale as Float)`	Change scaled game-time advancement.
`StartTimer(nodeId as String, waitTime = -1.0 as Float)`	Restart a Timer by node reference; an optional positive wait time overrides its interval.
`SetAnimationParameter(nodeId as String, name as String, value as Dynamic)`	Set an initialized graph parameter on a node, including triggers set to true.
`StopTimer(nodeId as String)`	Stop the referenced Timer.
`SpawnFromSpawner(nodeId as String) as Dynamic`	Request a manual spawn/acquisition from a Spawner2D and receive its result.
`MarkUIDirty()`	Invalidate UI rendering state after low-level changes.
`GetFocusedNodeId() as String`	Return the currently focused UI node ID.
`GetFocusedNode() as Dynamic`	Return the currently focused UI node or invalid.
`SetFocusedNode(nodeId as String)`	Assign remote focus to a suitable control.
`ChangeScreen(screenName as String)`	Replace the active UI screen.
`PushScreen(screenName as String)`	Overlay/push a UI screen.
`PopScreen() as Boolean`	Pop the current UI screen.
`ShowScreen(screenName as String)`	Show a named screen without replacing the screen stack.
`HideScreen(screenName as String)`	Hide a named screen.

Method signature	Purpose
`GetCurrentScreen() as String`	Query current UI screen.
`IsScreenVisible(screenName as String) as Boolean`	Query whether a named screen is visible.
`StartTransition(transType as String, options as Dynamic)`	Start a supported transition/preset with optional configuration.
`IsTransitionActive() as Boolean`	Query whether a transition is active.
`TransitionToScreen(screenName as String, options as Dynamic)`	Transition to a named UI screen.
`TransitionToScene(sceneName as String, options as Dynamic)`	Transition to a named runtime scene.
`SetLoadProgress(fraction as Float)`	Set progress used by the transition/loading workflow.
`MarkLoadComplete()`	Signal completion to the transition/loading workflow.
`IsInputBlocked() as Boolean`	Query whether transition state is currently blocking input.
`Storage_LoadFloat(namespace as String, key as String, defaultValue as Dynamic) as Float`	Read a float.
`Storage_LoadInt(namespace as String, key as String, defaultValue as Dynamic) as Integer`	Read an integer.
`Storage_LoadBool(namespace as String, key as String, defaultValue as Dynamic) as Boolean`	Read a boolean.
`Storage_Delete(namespace as String, key as String)`	Explicitly remove one stored key.
`Storage_ClearNamespace(namespace as String)`	Explicitly remove all keys in a namespace.
`Storage_Exists(namespace as String, key as String) as Boolean`	Test for a stored key.
`Storage_ListKeys(namespace as String) as Object`	List keys in a namespace.
`Options_Get(key as String) as String`	Read a defined option.
`Options_GetFloat(key as String) as Float`	Read a numeric option as float.
`Options_GetInt(key as String) as Integer`	Read a numeric option as integer.
`Options_GetBool(key as String) as Boolean`	Read a boolean option.
`Options_Set(key as String, value as Dynamic)`	Set a defined option value.

Method signature	Purpose
`Options_Reset(key as String)`	Reset one option to its default.
`Options_ResetAll()`	Reset all option defaults.
`Options_Cycle(key as String) as String`	Cycle through the option allowed values.
`Options_GetDisplayText(key as String) as String`	Return the option display representation.
`HighScores_Add(tableId as String, score as Integer, initials as String, extra as Dynamic) as Boolean`	Insert a score through the local table rules.
`HighScores_GetEntries(tableId as String) as Object`	Read local score entries.
`HighScores_Clear(tableId as String)`	Explicitly clear a local high-score table.
`HighScores_IsHighScore(tableId as String, score as Integer) as Boolean`	Test qualification for a table.
`HighScores_GetRowText(tableId as String, rank as Integer) as String`	Format a row for display.
`MarkNodeTransformDirty(nodeId as String)`	Invalidate cached transforms after low-level edits when the relevant path requires it.
`MarkSubtreeDirty(nodeId as String)`	Invalidate cached descendant transforms.
`GetWorldTransform(nodeId as String) as Object`	Read the node world transform.
`GetLocalTransform(nodeId as String) as Object`	Read the node local transform.
`ChangeScene(sceneName as String) as Boolean`	Change the active runtime scene through the scene-manager path.
`ReloadScene() as Boolean`	Reload a fresh authored copy of the current scene.
`PushScene(sceneName as String) as Boolean`	Push scene-stack state and change to another scene.
`PopScene() as Boolean`	Return through the scene-stack workflow.
`FindNodeByName(nodeName as String) as Dynamic`	Find a node in the active scene by name; invalid means not found. Stable IDs avoid duplicate-name ambiguity.
`ShakeCamera(magnitude as Float, duration as Float)`	Request shake with strength, duration and optional frequency; camera settings must permit shake.

Method signature	Purpose
`QuerySegment(x1 as Float, y1 as Float, x2 as Float, y2 as Float, radius = 0.0 as Float, skipId = "" as String, colliders = invalid as Dynamic) as Object`	Query a world-space swept point/projectile against native box or circle contacts. Returns nearest-first hits with nodeId, fraction, x, y and collider. Optional radius expands the target, skipId excludes a node, and colliders reuses a snapshot.
`ControlAudio(nodeReference as String, command as String, componentId = "" as String) as Boolean`	Send play/pause/resume/stop to a source; one-shots support play/stop. Optional component ID chooses one of several sources.
`DestroyNode(nodeId as String)`	Remove a runtime node/subtree through engine cleanup. Pool objects should normally use PoolRelease.
`PoolAcquire(poolName as String, x as Float, y as Float) as Dynamic`	Acquire an available item from a named pool at x/y; returns invalid when exhausted.
`PoolRelease(nodeId as String)`	Release/suspend a pooled node by ID.
`PoolReleaseAll(poolName as String)`	Release active items in one pool.
`PoolReleaseAllPools()`	Release items in all pools.
`PoolActiveCount(poolName as String) as Integer`	Return the number of acquired items in a named pool.
`PoolGetActive(poolName as String) as Object`	Return active pool items for iteration.

Advanced engine-bound signature index

The remaining bound methods are implementation/extension interfaces. Their names and source files are supplied for source navigation, not as stable beginner gameplay APIs. Calling render, cache, input-pump or phase functions out of order can break the engine. Prefer the game-facing operations above.

Bound signature	Canonical implementation
`Run()`	`RokuEngine_Run` source/engine/core.brs
`RunFrame(rawDt as Float)`	`RokuEngine_RunFrame` source/engine/core.brs
`BeginFrameTiming(rawDt as Float)`	`RokuEngine_BeginFrameTiming` source/engine/core.brs
`EndPhaseTiming(fieldName as String)`	`RokuEngine_EndPhaseTiming` source/engine/core.brs

Bound signature	Canonical implementation
`ResetSceneFrameStats()`	`RokuEngine_ResetSceneFrameStats` source/engine/core.brs
`CreateEmptyPhaseLists() as Object`	`RokuEngine_CreateEmptyPhaseLists` source/engine/core.brs
`MarkPhaseCachesDirty()`	`RokuEngine_MarkPhaseCachesDirty` source/engine/core.brs
`EnsurePhaseCaches()`	`RokuEngine_EnsurePhaseCaches` source/engine/core.brs
`RebuildPhaseCaches()`	`RokuEngine_RebuildPhaseCaches` source/engine/core.brs
`BuildTraversalNodeIds() as Object`	`RokuEngine_BuildTraversalNodeIds` source/engine/core.brs
`CreateComponentRegistry() as Object`	`RokuEngine_CreateComponentRegistry` source/engine/component_registry.brs
`RegisterBuiltinComponents(registry as Object)`	`RokuEngine_RegisterBuiltinComponents` source/engine/component_registry.brs
`RegisterComponentDefinition(definition as Object)`	`RokuEngine_RegisterComponentDefinition` source/engine/component_registry.brs
`GetComponentDefinition(typeName as Dynamic) as Dynamic`	`RokuEngine_GetComponentDefinition` source/engine/component_registry.brs
`GetComponentPhaseMetadata(typeName as Dynamic) as Dynamic`	`RokuEngine_GetComponentPhaseMetadata` source/engine/component_registry.brs
`ResolveComponentPriority(component as Object, phaseName as String, defaultPriority as Integer) as Integer`	`RokuEngine_ResolveComponentPriority` source/engine/core.brs
`AddComponentPhaseEntry(phaseLists as Object, entry as Object)`	`RokuEngine_AddComponentPhaseEntry` source/engine/core.brs
`SortPhaseEntries(entries as Object) as Object`	`RokuEngine_SortPhaseEntries` source/engine/core.brs
`RunComponentPhase(phaseName as String, dt as Float, event as Dynamic)`	`RokuEngine_RunComponentPhase` source/engine/core.brs
`DispatchComponentPhaseEntry(entry as Object, dt as Float, event as Dynamic)`	`RokuEngine_DispatchComponentPhaseEntry` source/engine/core.brs

Bound signature	Canonical implementation
`EnsureScriptReady(node as Object, component as Object, dt as Float)`	`RokuEngine_EnsureScriptReady` source/engine/core.brs
`PumpPlatformMessages()`	`RokuEngine_PumpPlatformMessages` source/engine/input.brs
`ProcessMessages(msg as Dynamic)`	`RokuEngine_ProcessMessages` source/engine/input.brs
`StageInputEvents()`	`RokuEngine_StageInputEvents` source/engine/core.brs
`ProcessInputPhase(dt as Float)`	`RokuEngine_ProcessInputPhase` source/engine/core.brs
`RunPreUpdatePhase()`	`RokuEngine_RunPreUpdatePhase` source/engine/core.brs
`RunFixedStepLoop()`	`RokuEngine_RunFixedStepLoop` source/engine/core.brs
`RunFixedStep(dt as Float)`	`RokuEngine_RunFixedStep` source/engine/core.brs
`RunVariableUpdatePhase()`	`RokuEngine_RunVariableUpdatePhase` source/engine/core.brs
`RunLateUpdatePhase()`	`RokuEngine_RunLateUpdatePhase` source/engine/core.brs
`FinalizeCollisionPhase()`	`RokuEngine_FinalizeCollisionPhase` source/engine/core.brs
`SyncRenderState()`	`RokuEngine_SyncRenderState` source/engine/core.brs
`ApplyPendingSceneChange()`	`RokuEngine_ApplyPendingSceneChange` source/engine/scene_manager.brs
`LoadScene(sceneName as String) as Object`	`RokuEngine_LoadScene` source/engine/scene_manager.brs
`Update(dt as Float)`	`RokuEngine_Update` source/engine/animation.brs
`UpdateComponents(dt as Float)`	`RokuEngine_UpdateComponents` source/engine/core.brs

Bound signature	Canonical implementation
`ProcessNodeComponents(node as Object, phase as String, dt as Float, event as Dynamic)`	`RokuEngine_ProcessNodeComponents` source/engine/core.brs
`ProcessComponentPhaseEntry(node as Object, component as Object, phase as String, dt as Float, event as Dynamic)`	`RokuEngine_ProcessComponentPhaseEntry` source/engine/core.brs
`InvokeComponentPhase(node as Object, component as Object, phase as String, dt as Float, event as Dynamic)`	`RokuEngine_InvokeComponentPhase` source/engine/component_registry.brs
`UpdateMovementComponents(dt as Float)`	`RokuEngine_UpdateMovementComponents` source/engine/core.brs
`UpdatePhysicsComponents(dt as Float)`	`RokuEngine_UpdatePhysicsComponents` source/engine/core.brs
`UpdateInputComponents(dt as Float)`	`RokuEngine_UpdateInputComponents` source/engine/core.brs
`UpdateAnimationComponents(dt as Float)`	`RokuEngine_UpdateAnimationComponents` source/engine/core.brs
`RunCollisionPass(dt as Float)`	`RokuEngine_RunCollisionPass` source/engine/core.brs
`RunScriptHooks(dt as Float, event as Dynamic)`	`RokuEngine_RunScriptHooks` source/engine/core.brs
`UpdateAudioComponents()`	`RokuEngine_UpdateAudioComponents` source/engine/core.brs
`Render()`	`RokuEngine_Render` source/engine/render_server.brs
`MarkUIClean()`	`RokuEngine_MarkUIClean` source/engine/render_server.brs
`GetRenderTraversalNodeIds() as Object`	`RokuEngine_GetRenderTraversalNodeIds` source/engine/render_server.brs
`RenderPassWorldSprites()`	`RokuEngine_RenderPassWorldSprites` source/engine/render_server.brs
`RenderPassTileMaps(onlyNodeId = "" as String)`	`RokuEngine_RenderPassTileMaps` source/engine/renderers/sprite_renderer.brs

Bound signature	Canonical implementation
`RenderPassWorldDebug()`	`RokuEngine_RenderPassWorldDebug` source/engine/render_server.brs
`RenderPassUI()`	`RokuEngine_RenderPassUI` source/engine/render_server.brs
`RenderPassOverlay()`	`RokuEngine_RenderPassOverlay` source/engine/render_server.brs
`RenderUiPass()`	`RokuEngine_RenderUiPass` source/engine/renderers/ui_renderer.brs
`RenderUiNode(node as Object, screenTransform as Object)`	`RokuEngine_RenderUiNode` source/engine/renderers/ui_renderer.brs
`DrawUiText(textValue as Dynamic, x as Integer, y as Integer, colorHex as Dynamic)`	`RokuEngine_DrawUiText` source/engine/renderers/ui_renderer.brs
`DrawUiTextAligned(textValue as Dynamic, x as Integer, y as Integer, w as Integer, h as Integer, colorHex as Dynamic, fontSize as Dynamic, align as Dynamic, valign as Dynamic, opacity as Dynamic, textStyle = invalid as Dynamic)`	`RokuEngine_DrawUiTextAligned` source/engine/renderers/ui_renderer.brs
`RenderDebugNode(node as Object, world as Object)`	`RokuEngine_RenderDebugNode` source/engine/renderers/debug_renderer.brs
`HandleInputEvent(msg as Object)`	`RokuEngine_HandleInputEvent` source/engine/input.brs
`HandleButtonCode(code as Integer, isPress as Dynamic, isRelease as Dynamic)`	`RokuEngine_HandleButtonCode` source/engine/input.brs
`InitializeUiRuntime()`	`RokuEngine_InitializeUiRuntime` source/engine/ui_system.brs
`RebuildUiRuntime()`	`RokuEngine_RebuildUiRuntime` source/engine/ui_system.brs
`GetUiTraversalNodeIds() as Object`	`RokuEngine_GetUiTraversalNodeIds` source/engine/ui_system.brs
`BuildUiTraversalNodeIds() as Object`	`RokuEngine_BuildUiTraversalNodeIds` source/engine/ui_system.brs
`GetNodeScreenTransform(nodeId as String) as Object`	`RokuEngine_GetNodeScreenTransform` source/engine/ui_system.brs

Bound signature	Canonical implementation
`IsUiNodeType(nodeType as Dynamic) as Boolean`	`RokuEngine_IsUiNodeType` source/engine/ui_system.brs
`IsWorldNodeType(nodeType as Dynamic) as Boolean`	`RokuEngine_IsWorldNodeType` source/engine/ui_system.brs
`InitializeFocusManager()`	`RokuEngine_InitializeFocusManager` source/engine/ui/ui_focus_manager.brs
`RebuildFocusables()`	`RokuEngine_RebuildFocusables` source/engine/ui/ui_focus_manager.brs
`MoveFocus(directionStep as Integer)`	`RokuEngine_MoveFocus` source/engine/ui/ui_focus_manager.brs
`MoveFocusDirection(direction as String)`	`RokuEngine_MoveFocusDirection` source/engine/ui/ui_focus_manager.brs
`HandleFocusInput(event as Object)`	`RokuEngine_HandleFocusInput` source/engine/ui/ui_focus_manager.brs
`UpdateFocusState()`	`RokuEngine_UpdateFocusState` source/engine/ui/ui_focus_manager.brs
`IsNodeFocusable(node as Object) as Boolean`	`RokuEngine_IsNodeFocusable` source/engine/ui/ui_focus_manager.brs
`HandleComponentInput(code as Integer)`	`RokuEngine_HandleComponentInput` source/engine/core.brs
`GetResourceManifest() as Dynamic`	`RokuEngine_GetResourceManifest` source/engine/core.brs
`InitializeScreenManager()`	`RokuEngine_InitializeScreenManager` source/engine/ui/ui_screen_manager.brs
`LoadScreensFromManifest()`	`RokuEngine_LoadScreensFromManifest` source/engine/ui/ui_screen_manager.brs
`ShowStartupScreen()`	`RokuEngine_ShowStartupScreen` source/engine/ui/ui_screen_manager.brs
`SetFocusToScreen(screenName as String)`	`RokuEngine_SetFocusToScreen` source/engine/ui/ui_screen_manager.brs
`InitializeTransitionManager()`	`RokuEngine_InitializeTransitionManager` source/engine/ui/ui_transition_manager.brs

Bound signature	Canonical implementation
`UpdateTransition(dt as Float)`	`RokuEngine_UpdateTransition` source/engine/ui/ui_transition_manager.brs
`RenderTransitionOverlay()`	`RokuEngine_RenderTransitionOverlay` source/engine/ui/ui_transition_manager.brs
`InitializeStorage()`	`RokuEngine_InitializeStorage` source/engine/ui/ui_storage.brs
`Storage_Save(namespace as String, key as String, value as Dynamic)`	`RGE_Storage_Save` source/engine/ui/ui_storage.brs
`Storage_Load(namespace as String, key as String, defaultValue as Dynamic) as String`	`RGE_Storage_Load` source/engine/ui/ui_storage.brs
`InitializeOptions()`	`RokuEngine_InitializeOptions` source/engine/ui/ui_options.brs
`UpdateOptionBindings()`	`RokuEngine_UpdateOptionBindings` source/engine/ui/ui_options.brs
`InitializeHighScores()`	`RokuEngine_InitializeHighScores` source/engine/ui/ui_high_scores.brs
`UpdateHighScoreBindings()`	`RokuEngine_UpdateHighScoreBindings` source/engine/ui/ui_high_scores.brs
`UpdateUiBindings()`	`RokuEngine_UpdateUiBindings` source/engine/ui_system.brs
`HandleAudioEvent(msg as Object)`	`RokuEngine_HandleAudioEvent` source/engine/core.brs
`DrawNodeTree()`	`RokuEngine_DrawNodeTree` source/engine/render_server.brs
`DrawNode(nodeId as String)`	`RokuEngine_DrawNode` source/engine/render_server.brs
`DrawNodeShapes(node as Object, world as Object)`	`RokuEngine_DrawNodeShapes` source/engine/render_server.brs
`SyncCompositorSprites()`	`RokuEngine_SyncCompositorSprites` source/engine/renderers/sprite_renderer.brs
`SyncNodeSprite(nodeId as String, active as Object, layer as Integer)`	`RokuEngine_SyncNodeSprite` source/engine/renderers/sprite_renderer.brs

Bound signature	Canonical implementation
`EnsureSpriteNodeRuntime(node as Object) as Object`	`RokuEngine_EnsureSpriteNodeRuntime` source/engine/core.brs
`RemoveNodeCompositorSprite(node as Object)`	`RokuEngine_RemoveNodeCompositorSprite` source/engine/core.brs
`RemoveAllCompositorSprites()`	`RokuEngine_RemoveAllCompositorSprites` source/engine/core.brs
`ApplySpriteTransform(node as Object, world as Object, spriteState as Object, drawX as Integer, drawY as Integer)`	`RokuEngine_ApplySpriteTransform` source/engine/renderers/sprite_renderer.brs
`ComputeSpriteScale(node as Object, world as Object, spriteState as Object) as Object`	`RokuEngine_ComputeSpriteScale` source/engine/renderers/sprite_renderer.brs
`CreateIdentityTransform() as Object`	`RokuEngine_CreateIdentityTransform` source/engine/transform_system.brs
`EnsureTransformRuntime(scene as Object)`	`RokuEngine_EnsureTransformRuntime` source/engine/transform_system.brs
`UpdateWorldTransforms(scene as Object)`	`RokuEngine_UpdateWorldTransforms` source/engine/transform_system.brs
`CreateResourceManager(engine as Object) as Object`	`RokuEngine_CreateResourceManager` source/engine/resource_manager.brs
`ComputeWorldTransform(node as Object, parentTransform as Object) as Object`	`RokuEngine_ComputeWorldTransform` source/engine/transform_system.brs
`BuildNodeMap(nodes as Object) as Object`	`RokuEngine_BuildNodeMap` source/engine/scene_manager.brs
`LoadSceneAsset(path as String) as Dynamic`	`RokuEngine_LoadSceneAsset` source/engine/scene_manager.brs
`LoadImageAsset(resourcePath as Dynamic) as Dynamic`	`RokuEngine_LoadImageAsset` source/engine/renderers/sprite_renderer.brs
`LoadSpriteSheetTile(basePath as String, tileIndex as Integer) as Dynamic`	`RokuEngine_LoadSpriteSheetTile` source/engine/renderers/sprite_renderer.brs
`ParseTileAssetPath(resourcePath as Dynamic) as Dynamic`	`RokuEngine_ParseTileAssetPath` source/engine/renderers/sprite_renderer.brs
`FindSpriteSheetByResourcePath(basePath as String) as Dynamic`	`RokuEngine_FindSpriteSheetByResourcePath` source/engine/renderers/sprite_renderer.brs

Bound signature	Canonical implementation
`BuildAnimationFrames(node as Object) as Object`	`RokuEngine_BuildAnimationFrames` source/engine/animation.brs
`ComputeAnimationFrameIndex(node as Object, frameCount as Integer) as Integer`	`RokuEngine_ComputeAnimationFrameIndex` source/engine/animation.brs
`DrawDebugText()`	`RokuEngine_DrawDebugText` source/engine/renderers/debug_renderer.brs
`RootNodeIds() as Object`	`RokuEngine_RootNodeIds` source/engine/scene_manager.brs
`RefreshSceneCaches()`	`RokuEngine_RefreshSceneCaches` source/engine/scene_manager.brs
`BuildChildrenMap(nodes as Object) as Object`	`RokuEngine_BuildChildrenMap` source/engine/scene_manager.brs
`BuildNamedNodeMap(nodes as Object) as Object`	`RokuEngine_BuildNamedNodeMap` source/engine/scene_manager.brs
`UpdateActiveCameraNode()`	`RokuEngine_UpdateActiveCameraNode` source/engine/scene_manager.brs
`ApplyCameraFollow(dt as Float)`	`RokuEngine_ApplyCameraFollow` source/engine/camera_policies.brs
`EnsureNodeRuntime(node as Object) as Object`	`RokuEngine_EnsureNodeRuntime` source/engine/core.brs
`GetNodeCollider(node as Object) as Dynamic`	`RokuEngine_GetNodeCollider` source/engine/core.brs
`BuildColliderList() as Object`	`RokuEngine_BuildColliderList` source/engine/core.brs
`ApplyRigidBodyMotion(node as Object, component as Object, dt as Float)`	`RokuEngine_ApplyRigidBodyMotion` source/engine/core.brs
`ResolveRigidBodyCollisions(node as Object, component as Object, colliders as Object)`	`RokuEngine_ResolveRigidBodyCollisions` source/engine/core.brs
`OverlapsCollider(target as Object, colliders as Object, skipNodeId as String) as Dynamic`	`RokuEngine_OverlapsCollider` source/engine/core.brs
`ProcessCollisions()`	`RokuEngine_ProcessCollisions` source/engine/core.brs

Bound signature	Canonical implementation
`PlayNodeAudio(node as Object, component as Object)`	`RokuEngine_PlayNodeAudio` source/engine/core.brs
`InvokeScriptComponent(node as Object, component as Object, eventName as String, dt as Dynamic, extra as Dynamic)`	`RokuEngine_InvokeScriptComponent` source/engine/scripts.brs
`InvokeNodeScripts(node as Object, eventName as String, dt as Dynamic, extra as Dynamic)`	`RokuEngine_InvokeNodeScripts` source/engine/scripts.brs
`GetActionForButtonCode(code as Integer) as String`	`RokuEngine_GetActionForButtonCode` source/engine/input.brs
`SpawnNode(template as Object, parentId as Dynamic) as Dynamic`	`RokuEngine_SpawnNode` source/engine/core.brs
`GenerateId() as String`	`RokuEngine_GenerateId` source/engine/core.brs

Asset streamer interface

Method/property	Meaning
engine.assetStream.PreloadGroup(groupId, lazy)	Queue/request an existing manifest group. true enables incremental texture queue processing.
engine.assetStream.EnqueueLazy(guid, path, kind)	Queue an asset identified by GUID/path; use supported resource kinds.
engine.assetStream.IsLoaded(guid)	Query successful cache tracking; failed bitmap loads are not success.
engine.assetStream.budgetKb / batchSize / lastWarning	Bitmap budget / assets processed per frame / explanation when pinned content prevents eviction.

27. Files, glossary and further reading

File/folder	Role
project.settings.json	Project name, input and display settings.
*.rkscene	Authored scene document.
*.rkscene.editstate	Editor resources, preview/workspace/resource state associated with the scene.
components/*.brs	Project behavior scripts with path-derived hook names.
images / assets / audio / fonts	Project source assets referenced by the build.
source/engine	Canonical native engine runtime in this repository.
.visual-editor/engine-builds	Generated build output, replaced by subsequent builds.
editor-preferences.json in Electron userData	Application-wide theme preference, separate from project data.
samples/collect-the-stars	Editable first-game example used in screenshots.
samples/gameplay-helpers	Gameplay helper nodes and event-hook examples.

Glossary: AABB: axis-aligned bounding box. Atlas: regions sharing a texture. Bake: pre-render processing into a playback asset. Collider: bounds used for interactions. D-pad: directional remote control. Easing: transition timing curve. Focus: control receiving remote navigation. Gizmo: editing guide. LRU: least-recently-used eviction. Nine-slice: scalable texture preserving borders. Pool: preallocated reusable objects. Prefab/instance: reusable scene content. Sidecar: associated editor-state file. Sideload: developer-channel install. Telemetry: runtime-reported measurements/estimates.

Further repository guides: README.md, docs/SCRIPTING.md, docs/SPRITE_SLICING.md, docs/gameplay-nodes.md, docs/NATIVE_CONTROLS.md, docs/native-systems.md and docs/system-audit.md. For current implementation details, prefer canonical source and the latest audit over historical architecture notes.